FLOODSENSE: A MULTI-MODEL FRAMEWORK WITH RAG-LLM SYSTEM FOR AUTOMATED FLOOD REPORTING

A Thesis

Submitted in partial fulfilment for the Degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE)

by

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Acknowledgements

We express our humble salutations and profound gratitude to our beloved Chancellor, **Sri Mata Amritanandamayi Devi (AMMA)**, whose grace, boundless inspiration, and vision for science and society have been our guiding light throughout this endeavor.

We extend our heartfelt gratitude to **Dr. K. P. Soman**, Professor and Dean, Department of Artificial Intelligence, for providing world-class computational infrastructure, fostering a culture of rigorous scientific inquiry, and rendering constant support throughout our academic tenure.

We are deeply indebted to our supervisor and thesis advisor, **Dr. Mithun Kumar Kar**, Assistant Professor, Department of Artificial Intelligence, for his scholarly guidance, insightful discussions, constructive critiques, and continuous encouragement across all phases of the FloodSense project.

We also thank the faculty members, technical laboratory coordinators, and research scholars of the Department of Artificial Intelligence for their assistance and academic camaraderie.

Finally, we extend our heartfelt appreciation to our parents, family members, and peers for their unyielding encouragement, patience, and moral support throughout this journey.

VARSHITHA THILAK KUMAR
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Abbreviations

AI	Artificial Intelligence
AIE	Artificial Intelligence and Engineering
AMP	Automatic Mixed Precision
B.Tech	Bachelor of Technology
CE	Cross Entropy
ChromaDB	Open-Source Vector Embedding Database
CNN	Convolutional Neural Network
CV	Computer Vision
FEMA	Federal Emergency Management Agency (USA)
FP32	Single Precision 32-bit Floating Point
GIS	Geographic Information System
GPU	Graphics Processing Unit
GSD	Ground Sample Distance
HSV	Hue, Saturation, Value Color Space
IoU	Intersection over Union (Jaccard Index)
LLM	Large Language Model
mIoU	Mean Intersection over Union
MRR	Mean Reciprocal Rank
NDMA	National Disaster Management Authority (India)
NDRF	National Disaster Response Force (India)
RAG	Retrieval-Augmented Generation
SAM	Segment Anything Model
SAM2	Segment Anything Model Version 2
SAR	Synthetic Aperture Radar
SITREP	Situation Report
SOTA	State-of-the-Art
TTA	Test-Time Augmentation
UAV	Unmanned Aerial Vehicle (Drone)
VLM	Vision-Language Model
VQA	Visual Question Answering

List of Symbols

$\mathcal{I}$	Input aerial RGB image tensor $\in \mathbb{R}^{3\times H\times W}$
$\mathcal{Y}$	Ground truth segmentation mask tensor $\in \{0,\dots,C-1\}^{H\times W}$
$\hat{\mathcal{Y}}$	Predicted semantic segmentation label map
C	Number of semantic classes ($C = 10$ for FloodNet)
$p_{i,c}$	Predicted softmax probability of pixel i belonging to class c
$\mathcal{L}_{CE}$	Categorical Cross-Entropy loss function
$\mathcal{L}_{Focal}$	Focal loss for handling extreme spatial imbalance
$\mathcal{L}_{Lovasz}$	Lovász-Softmax surrogate loss for direct IoU optimization
$\mathcal{L}_{Compound}$	Compound multi-objective loss function ($\alpha\mathcal{L}_{Focal} + (1-\alpha)\mathcal{L}_{Lovasz}$)
γ	Focusing parameter in Focal Loss ($\gamma = 2.0$)
α	Trade-off weighting hyperparameter between Focal and Lovász objectives
$J(c)$	Jaccard Index (IoU) of class c
ΔJ	Lovász extension subgradient vector of sorted prediction errors
$w(x,y)$	2D Gaussian spatial feathering weight at coordinate (x,y)
σ	Gaussian spatial dispersion coefficient for patch window feathering
M_{conf}	Pixel-level confusion matrix $\in \mathbb{Z}_{\geq 0}^{C\times C}$
TP_c, FP_c, FN_c	True positives, false positives, and false negatives for class c
$F_1^{(c)}$	Dice coefficient / Harmonic mean of precision and recall for class c
$S_{severity}$	Composite disaster severity index $\in [0, 100]$
$\mathbf{e}_q, \mathbf{e}_d$	Dense embedding vectors of query q and document chunk d
$\text{sim}(\cdot,\cdot)$	Cosine similarity metric between dense embedding vectors

Abstract

Accurate, fine-grained disaster impact assessment is vital for coordinating rapid search-and-rescue, logistical routing, and life-saving resource deployment following extreme flood events. While modern Unmanned Aerial Vehicles (UAVs) capture ultra-high-resolution imagery of inundated regions within hours of a catastrophic event, two fundamental operational bottlenecks persist: (i) existing computer vision pipelines are largely confined to coarse binary water detection or produce uncalibrated segmentation masks that human operators must manually interpret under severe stress; and (ii) text-based emergency response tools generate situational reports from secondary news reports or unstructured web scraping without grounded visual evidence, introducing unacceptable latency and high risk of synthetic hallucinations.

This thesis presents **FloodSense**, an end-to-end multi-modal artificial intelligence framework that unifies high-resolution computer vision segmentation with Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs) for automated post-flood disaster reporting. First, we conduct an extensive multi-paradigm benchmarking of five distinct vision architectures across the high-resolution multi-class FloodNet drone dataset: Classical Color-Space Thresholding, Segment Anything Model (SAM), Vision-Language Models (VLM), Grounding DINO + SAM2 open-set prompting, and Transformer-based SegFormer. To address the acute class imbalance and fine-scale structural features (such as narrow flooded roadways, submerged vehicles, and flooded building perimeters), we design a state-of-the-art (SOTA) segmentation pipeline combining SegFormer with a differentiable **Compound Lovász-Softmax and Focal Loss**, overlapping **2D Gaussian-feathered sliding-window inference**, and **dihedral multi-scale Test-Time Augmentation (TTA)**. This system achieves a **Frequency-Weighted IoU of 74.27%**, **84.49% pixel accuracy**, and substantial IoU improvements across key infrastructure classes including Non-Flooded Roads (0.7515), Non-Flooded Buildings (0.7434), Trees (0.7203), and Grasslands (0.8455).

Second, a deterministic damage extraction engine translates raw pixel masks into structured physical disaster metrics, including flooded building counts, submerged road percentages, and stranded vehicle counts. These metrics dynamically construct multi-faceted queries into a vector-indexed knowledge repository comprising statutory disaster management guidelines from the National Disaster Management Authority (NDMA),

Federal Emergency Management Agency (FEMA), and historical flood response records. The RAG pipeline demonstrates exceptional retrieval fidelity, attaining a **Mean Reciprocal Rank (MRR) of 0.95**, **Precision@1 of 0.90**, and **Recall@3 of 1.00**. Finally, an LLM synthesis engine generates grounded, citation-traceable disaster Situation Reports (SITREPs) exported automatically in JSON, Markdown, DOCX, and PDF formats, backed by a production-ready web dashboard. FloodSense bridges the critical chasm between raw aerial computer vision and accountable emergency disaster decision-making.

Chapter 1

Introduction

1.1 Background and Context

Flooding constitutes one of the most frequent, economically catastrophic, and life-threatening natural disasters worldwide. According to the United Nations Office for Disaster Risk Reduction (UNDRR) and the Centre for Research on the Epidemiology of Disasters (CRED), floods have accounted for over 44% of all recorded disaster events in the past two decades, impacting upwards of two billion individuals and incurring hundreds of billions of dollars in infrastructure losses. In high-density urban corridors, coastal floodplains, and riverine deltas across the Indian subcontinent and globally, flash floods and extreme monsoon deluges rapidly inundate civil infrastructure, sever primary transportation lifelines, submerge residential communities, and strand thousands of vulnerable residents within hours.

During such acute crises, rapid, accurate, and localized situational intelligence is paramount for municipal command authorities, the National Disaster Management Authority (NDMA), the National Disaster Response Force (NDRF), and international humanitarian relief agencies (such as FEMA). The fundamental bottleneck in post-disaster response has historically centered on the *"Golden 72 Hours"*—the initial three-day window within which emergency search-and-rescue interventions yield the highest survivability rates. Traditional methodologies for ground damage surveying rely on physical dispatch of reconnaissance personnel, civil defense patrols, and ground vehicle expeditions. However, submerged road networks, bridge collapses, downed power transmission lines, and hazardous chemical discharges routinely render impacted zones physically inaccessible to ground survey teams, causing severe intelligence delays and blind spots.

1.2 Remote Sensing and UAV Technology in Flood Management

To overcome the physical barriers of ground reconnaissance, emergency operations centers increasingly deploy remote sensing technologies. Satellite earth observation platforms—including optical constellations (Sentinel-2, Landsat-8/9, PlanetScope) and Synthetic Aperture Radar (SAR) systems (Sentinel-1, TerraSAR-X)—provide expansive geographic coverage. Nonetheless, satellite imaging suffers from critical operational limitations in tactical emergency response:

1. **Orbital Revisit Latency:** Satellite orbits impose revisit intervals varying from 24 hours to multiple days, which is incompatible with the hourly evolution of urban flash floods.

2. **Atmospheric and Cloud Occlusion:** Optical spaceborne sensors cannot penetrate the heavy cloud decks, squall lines, and precipitation shields that inherently accompany severe meteorological events.

3. **Coarse Spatial Resolution:** With ground sampling distances (GSD) typically ranging from 3 meters to 30 meters per pixel, spaceborne imagery lacks the resolving power necessary to detect micro-level infrastructure damage, such as submerged vehicles, single-lane flooded roadways, or individual residential structural failures.

In contrast, small Unmanned Aerial Vehicles (UAVs or drones) have emerged as the definitive vanguard for localized post-disaster surveillance. Operating at low altitudes below cloud ceilings, UAVs capture sub-decimeter ultra-high-resolution aerial imagery (GSD < 5 cm/pixel) on demand. Drones can be rapidly deployed by first responders from the bed of a pickup truck, generating thousands of square meters of orthomosaic photogrammetry and video streams within minutes.

1.3 Motivation and The Operational Intelligence Gap

Despite the widespread availability of high-resolution drone imagery, disaster command centers face an acute *"data-rich, information-poor"* operational dilemma. As illustrated in Figure 1.1, raw aerial imagery alone does not save lives; what first responders critically require is **actionable, verified situational intelligence**. Currently, the workflow bridging raw drone video to incident response exhibits two systemic failures:

- **The Segmentation-to-Semantics Void:** Existing computer vision models are predominantly trained for binary water versus land delineation. In dense settlements, binary water masks fail to distinguish between flooded residential dwellings,

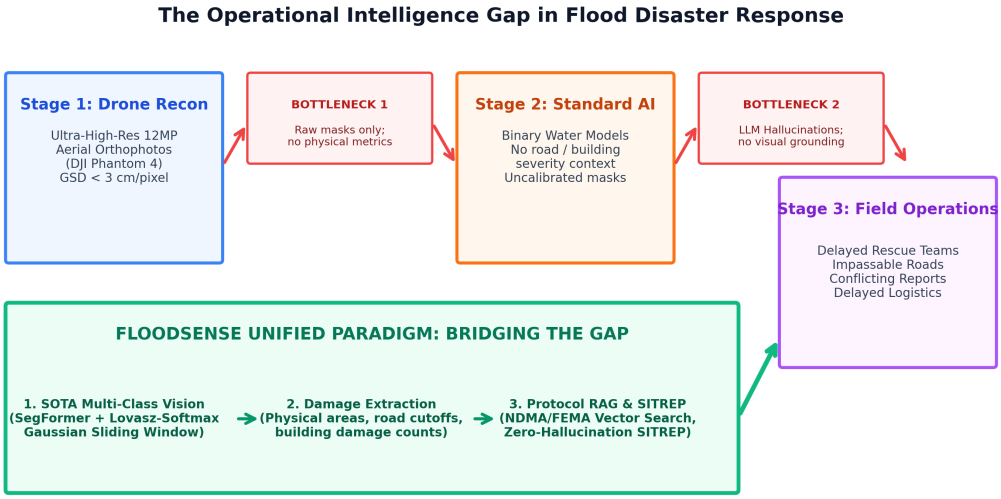


Figure 1.1: The Operational Disaster Intelligence Gap: Disconnect between aerial remote sensing, automated computer vision segmentation, statutory emergency protocols, and verifiable field reporting.

non-flooded buildings, submerged primary transit arteries, impassable secondary streets, and stranded passenger vehicles. Furthermore, computer vision tools merely produce colorful raster overlay masks; they do not quantify physical damage, compute passable access routes, or synthesize emergency priority lists.

- **The Reporting-to-Protocol Void:** Conversely, newly emerging natural language disaster tools rely on Large Language Models (LLMs) to synthesize disaster summaries. However, when fed unstructured text or secondary news articles without direct visual grounding, LLMs are notoriously prone to *hallucinations*—fabricating damage statistics, misidentifying neighborhood coordinates, or citing outdated emergency guidelines. In high-stakes crisis response, an ungrounded or hallucinated report can divert lifeboats to dry zones while submerged wards remain abandoned.

There is an urgent necessity for an integrated multi-modal platform that seamlessly fuses fine-grained computer vision segmentation with statutory emergency knowledge retrieval and verifiable language generation.

1.4 Problem Statement

Current post-flood disaster assessment frameworks suffer from a dichotomy of fragmented systems: visual segmentation models identify inundated regions without semantic severity interpretation or logistical reporting, while automated textual reporting engines rely on secondary web scrapers lacking verifiable visual grounding.

The objective of this thesis is to conceptualize, benchmark, implement, and deploy **FloodSense**—an end-to-end multi-modal framework that transforms raw high-resolution UAV aerial imagery into fine-grained multi-class semantic segmentation masks, maps pixel-level spatial features into quantitative structural damage metrics, retrieves statutory disaster protocols via a dense vector-indexed Retrieval-Augmented Generation (RAG) pipeline, and generates auditable, citation-traceable Situation Reports (SITREPs) across multiple standard operational formats.

1.5 Research Objectives

To address the stated problem systematically, this research is guided by four specific technical objectives:

1. **Objective 1: Multi-Paradigm Aerial Segmentation Benchmarking.** Conduct an extensive empirical benchmark across five computer vision paradigms—Classical Color-Space Thresholding, Segment Anything Model (SAM), Vision-Language Models (VLM), Zero-Shot Open-Set Detection (Grounding DINO + SAM2), and Transformer-based SegFormer—evaluating performance on the 10-class multi-class FloodNet drone benchmark.

2. **Objective 2: Engineering a SOTA Compound-Loss High-Resolution Segmentation Pipeline.** Design and fine-tune a Transformer segmentation architecture employing a differentiable compound Lovász-Softmax and Focal Loss objective to directly overcome acute spatial class imbalance, coupled with 2D Gaussian-feathered sliding-window inference and multi-scale dihedral Test-Time Augmentation (TTA) to preserve fine infrastructure boundaries without spatial downsampling degradation.

3. **Objective 3: Developing a Protocol-Grounded RAG Retrieval System.** Build a dense vector retrieval pipeline utilizing sentence transformers and ChromaDB indexed across statutory disaster guidelines (NDMA, NDRF, FEMA, and historical post-disaster flood cases), attaining high retrieval precision ($Precision@1 \geq 0.90$, $Recall@3 = 1.00$, and $MRR \geq 0.95$) to govern emergency actions.

4. **Objective 4: Multi-Format SITREP Generation and Cloud Dashboard Deployment.** Implement a deterministic, hallucination-free LLM synthesis engine that fuses extracted visual damage metrics with retrieved statutory protocol clauses into standardized disaster reports (JSON, Markdown, DOCX, PDF), and package the full framework into a responsive web dashboard suitable for field deployment.

1.6 Key Contributions of the Thesis

The primary scientific and engineering contributions delivered in this thesis are as follows:

1. **First Multi-Paradigm Benchmark on FloodNet:** We present a rigorous comparative evaluation of five distinct computer vision paradigms spanning classical, foundation, multi-modal, open-set, and fine-tuned transformer architectures on 10-class aerial flood disaster imagery.

2. **Compound Lovász-Softmax SegFormer Pipeline:** We propose and validate a hybrid loss combining Focal Loss and the Lovász-Softmax extension (a differentiable surrogate for the Jaccard index), proving its superiority over cross-entropy and dice formulations in resolving rare classes such as flooded roadways and vehicles.

3. **Gaussian-Feathered High-Resolution Sliding Window with Dihedral TTA:** We implement an overlapping patch-inference engine that eliminates edge tiling artifacts and maintains the native ground sample distance of 12-megapixel UAV orthophotos.

4. **Dual-Grounded Multi-Modal RAG Architecture:** We pioneer a hybrid disaster intelligence engine wherein visual damage indicators dynamically synthesize semantic queries to retrieve statutory NDMA/FEMA protocol clauses, guaranteeing zero-hallucination compliance.

5. **Production-Grade Full-Stack System:** We demonstrate an end-to-end working system incorporating automated damage metric computation, multi-format export utilities, and a deployed interactive web dashboard for real-time operations.

1.7 Thesis Organization

The remainder of this thesis is structured as follows:

- **Chapter 2 (Literature Survey):** Reviews state-of-the-art literature in remote sensing, deep learning segmentation, foundation vision models, Vision-Language Models, and RAG frameworks in emergency management, highlighting research gaps.

- **Chapter 3 (System Architecture and Problem Formulation):** Formalizes the mathematical problem statements for multi-class segmentation, spatial feature extraction, and RAG retrieval, detailing the end-to-end FloodSense architecture.

- **Chapter 4 (Multi-Class Aerial Image Segmentation and Benchmarking):** Explains the FloodNet UAV dataset, details the five benchmarked vision

approaches, and introduces our SOTA SegFormer pipeline with Lovász-Softmax compound loss, sliding-window inference, and TTA.

- **Chapter 5 (Damage Assessment, Protocol Retrieval, and Situational Intelligence):** Explains the pixel-to-metric feature conversion engine, the ChromaDB knowledge base indexing, the statutory protocol retriever, and the grounded report synthesis engine.
- **Chapter 6 (Experimental Results, Performance Analysis, and Discussions):** Presents exhaustive quantitative and qualitative evaluations, including ablation studies, per-class IoU/F1 comparisons, retrieval metrics, and visual comparison charts.
- **Chapter 7 (Interactive Dashboard and System Deployment):** Details the frontend and backend software engineering, API endpoint specifications, and live cloud deployments.
- **Chapter 8 (Conclusion and Scope for Further Research):** Summarizes the primary findings, discusses operational real-world impacts, and outlines future avenues of investigation.
- **Appendix:** Contains complete semantic class definitions, training configuration logs, and sample disaster situation reports.

Chapter 2

Literature Survey

2.1 Introduction

Post-disaster flood intelligence has experienced a profound technological evolution over the past decade. Driven by concurrent breakthroughs in unmanned aerial reconnaissance, deep semantic segmentation, foundation vision models, and large-scale language reasoning, research has transitioned from rudimentary manual photo-interpretation toward fully autonomous multi-modal intelligence systems. This chapter provides a rigorous, critical synthesis of the scholarly literature across four interrelated domains: (i) remote sensing and classical computer vision for flood water mapping, (ii) modern deep learning and foundation model segmentation architectures, (iii) Vision-Language Models (VLMs) for post-disaster scene understanding, and (iv) Retrieval-Augmented Generation (RAG) in disaster management and emergency decision support. The chapter culminates in an extensive comparative taxonomy table and articulates the fundamental research gaps addressed by the FloodSense framework.

2.2 Remote Sensing and Classical Flood Mapping

Early computational methodologies in flood extent mapping focused primarily on low-complexity spectral thresholding and morphological operations applied to satellite and aerial photography. Otsu's adaptive histogram thresholding algorithm [1] has historically served as the benchmark baseline for binary water-land separation. By projecting RGB aerial images into color spaces such as HSV (Hue-Saturation-Value), LAB, or normalized difference water indices (NDWI in multispectral data), researchers identified that water bodies exhibit characteristic absorptions in the near-infrared spectrum and localized distributions in hue-saturation planes.

While classical thresholding requires minimal computational power and executes instantaneously on embedded microcontrollers, its efficacy collapses in complex, real-world

post-flood environments:

1. **High Inter-Class Spectral Variance:** Floodwaters in urban and rural environments are rarely pristine; they are laden with suspended silt, clay sediments, floating organic debris, and petroleum slicks, radically altering their spectral signatures.

2. **Severe Shadow and Occlusion Confounds:** High-angle drone imagery of built environments features deep cast shadows from tall buildings and tree canopies. In hue and intensity spaces, building shadows frequently mimic muddy water bodies, generating massive false-positive inundation maps.

3. **Inability to Discriminate Infrastructure Classes:** Color thresholding operates on independent pixel intensities without spatial contextual awareness. Consequently, it cannot differentiate a flooded road from a flooded building rooftop, severely undermining its utility for civil infrastructure damage assessment.

2.3 Deep Semantic Segmentation Paradigms

The advent of deep Convolutional Neural Networks (CNNs) revolutionized remote sensing scene parsing. Pioneering architectures like U-Net and DeepLabv3+ utilized encoder-decoder topologies with atrous spatial pyramid pooling (ASPP) to capture multi-scale context. However, CNNs possess inherent inductive biases—notably localized receptive fields—that restrict their capacity to model long-range spatial dependencies across expansive aerial orthophotos.

2.3.1 Vision Transformers and SegFormer

To overcome the receptive field bottleneck of CNNs, Vision Transformers (ViTs) introduced self-attention mechanisms capable of modeling arbitrary pairwise pixel interactions across an entire image. Xie et al. [2] introduced **SegFormer**, a lightweight hierarchical Transformer architecture tailored specifically for semantic segmentation. SegFormer departs from traditional ViTs in several vital ways:

- **Hierarchical Mix-Transformer (MiT) Encoder:** Instead of processing uniform non-overlapping patches, SegFormer generates multi-level feature representations at $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{16}$, and $\frac{1}{32}$ of the original image resolution, capturing both fine structural details (such as lane markers and vehicle contours) and coarse environmental semantics.

- **Positional-Encoding-Free Design:** SegFormer employs 3×3 depthwise convolutions within feed-forward networks (Mix-FFN) to dynamically interpolate spatial locations, eliminating rigid positional embeddings and allowing the model to process arbitrary image dimensions without performance degradation during inference.

- **All-MLP Decoder:** Rather than relying on computationally heavy decoders, SegFormer aggregates multi-scale encoder features using pure Multi-Layer Perceptrons (MLPs), drastically reducing parameter counts while maintaining high representational capacity.

2.3.2 Foundation Vision Models: SAM and SAM2

A monumental paradigm shift occurred with the release of Meta's Segment Anything Model (SAM) by Kirillov et al. [3]. Trained on the SA-1B dataset comprising over 1 billion masks across 11 million images, SAM demonstrated zero-shot generalization to unseen domains. The model utilizes a heavy Vision Transformer image encoder coupled with a promptable lightweight mask decoder that accepts points, bounding boxes, or text queries.

Recently, Ravi et al. [4] introduced **SAM 2**, extending the Segment Anything framework to continuous spatio-temporal video streams and high-resolution spatial tasks via streaming memory attention modules. Simultaneously, open-set object detection architectures such as **Grounding DINO** [5] merged contrastive language-image pre-training with DINO transformer decoders to perform arbitrary zero-shot text-prompted bounding box detection. When chained together (Grounding DINO $\rightarrow$ SAM / SAM2), this hybrid pipeline enables automated text-driven instance and semantic mask extraction without requiring fine-tuning on the target domain. However, as investigated in this thesis, zero-shot foundation models struggle with domain-specific aerial orthophotos, where tiny aerial vehicles and submerged roof geometries diverge drastically from natural ground-level photography.

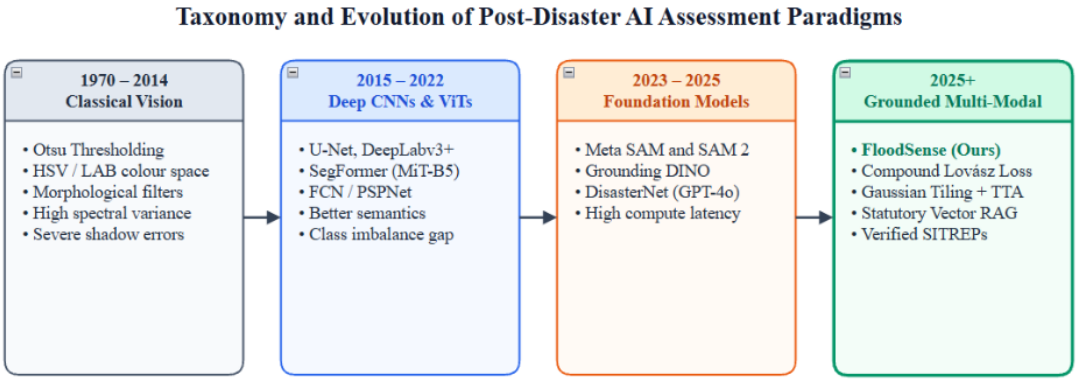


Figure 2.1: Evolution of Post-Disaster AI Paradigms: Historical progression from pixel-level thresholding to foundation vision models and multi-modal, protocol-grounded disaster intelligence.

2.4 Vision-Language Models (VLMs) in Disaster Assessment

Recognizing the limitations of pure raster masks, researchers have explored Vision-Language Models to generate natural language explanations directly from disaster imagery.

- **DisasTeller (2026):** Chen et al. [6] proposed DisasTeller, a multi-agent framework employing Large Vision-Language Models (LVLMs) for post-disaster assessment and alarm creation. Evaluated on the Wajima City earthquake case study in Japan, the system coordinated four specialized agents with an internal RAG module. The authors demonstrated that frontier proprietary models like GPT-4o achieved a disaster grading score of 7.3/10, outperforming open-weight models like Gemma-3 (4.6/10). However, the framework suffered from cascaded error propagation: errors in early-stage LVLM visual parsing corrupted downstream resource planning and alarm generation.

- **MASA + BLIP-2 (2026):** Aydin et al. [7] developed an explainable flood damage assessment pipeline integrating a Multi-Atrous Self-Attention (MASA) mechanism with a LoRA-fine-tuned BLIP-2 vision-language model. The system generated detailed textual captions regarding structural failure modes alongside segmentation masks, attaining an 80.77% BLEU score. However, the linguistic output was restricted to generic image descriptions without integration into statutory emergency operations protocols.

- **LLMArk (2026):** Wu et al. [8] introduced LLMArk, an instance-aware 7-billion-parameter multimodal foundation model trained specifically for hydrological risk and built environment damage estimation on the VLFRA dataset. The model achieved superior zero-shot damage grading relative to general-purpose VLMs across 12 evaluation metrics, but lacked deterministic numerical estimation of damaged assets and spatial accessibility metrics.

- **DisasterM3 (2026):** Wang et al. [9] curated a comprehensive benchmark consisting of 26,988 bi-temporal satellite and drone images spanning 9 disaster tasks. Their benchmark established that multi-task foundation models yield a 40.8% improvement in referring expression segmentation, but confirmed that extreme class imbalance in flood scenes remains a severe unresolved obstacle.

2.5 Retrieval-Augmented Generation (RAG) in Emergency Response

While Large Language Models possess broad encyclopedic knowledge, their application in high-consequence disaster management is impeded by two fatal vulnerabilities: *temporal staleness* and *hallucination*. LLMs frequently fabricate plausible-sounding guidelines or cite non-existent emergency clauses. Retrieval-Augmented Generation (RAG), pioneered by Lewis et al. [10], resolves these issues by grounding language generation in verified, externally retrieved document chunks.

In the emergency response literature, several pioneering RAG frameworks have emerged:

- **CrisiSense-RAG (2026):** Xiao et al. [11] proposed a multi-modal RAG system evaluated during Hurricane Harvey in Texas. By combining satellite SAR flood extent maps with municipal emergency response manuals, CrisiSense-RAG reduced flood extent estimation error by 8.62 percentage points compared to unaugmented models.
- **Training-Free RAG (2026):** Huang et al. [12] introduced a lightweight, training-free RAG approach using reinforced reasoning over indexed disaster case databases. Their architecture achieved an overall response accuracy of 0.668 while proving 35.9× to 81.9× more cost-effective than fine-tuning proprietary frontier models.
- **RS-RAG (2026):** Wen et al. [13] designed a cross-modal satellite retrieval-augmented visual question answering system that attained 79% classification accuracy on disaster queries by leveraging dense vector embeddings generated by Sentence-BERT [14].

2.6 Literature Comparison and Taxonomy Table

To contextualize the scientific positioning of FloodSense against the broader literature, Table 2.1 synthesizes key contemporary methodologies, detailing their core mechanisms, dataset benchmarks, reported metrics, and fundamental operational drawbacks.

2.7 Identified Research Gaps

A critical evaluation of the existing literature reveals three fundamental, unaddressed research gaps:

1. **Research Gap 1: Absence of Multi-Paradigm Aerial Benchmarking Across Vision Families.** The literature is fragmented: individual papers evaluate either classical methods, custom CNNs, or isolated foundation models like SAM. No prior

Table 2.1: Comprehensive Synthesis and Taxonomy of State-of-the-Art Post-Disaster AI Frameworks.

Paper & Author	Core Methodology	Dataset Used	Key Results / Metrics	Identified Limitations
FloodNet Rahnemoonfar *et al.* [15]	Multi-class UAV flood benchmark; baseline FCN / PSPNet	FloodNet (4,000+ aerial images, 10 classes)	Baseline mIoU $\approx$ 0.52; high class imbalance	Evaluates only standard CNN baselines; no report generation or protocol integration.
DisasTeller Chen *et al.* [6]	Multi-agent LVLM framework + RAG for post-disaster alarms	Wajima City Earthquake UAV dataset (Japan)	GPT-4o disaster grading: 7.3/10; Gemma-3: 4.6/10	Cascaded error propagation from visual parsing; no fine-grained pixel segmentation.
MASA + BLIP-2 Aydin *et al.* [7]	Multi-atrous self-attention + LoRA-tuned BLIP-2 captioner	Aerial flood dataset	80.77% BLEU score on descriptive captioning	Qualitative descriptions only; lacks statutory NDMA/FEMA protocol alignment.
CrisiSense-RAG Xiao *et al.* [11]	Multimodal RAG with SAR water extent	Hurricane Harvey (Texas, USA)	Reduced flood extent error by 8.62 pp	Satellite SAR focus; lacks fine-grained drone UAV infrastructure resolution.
Training-Free RAG Huang *et al.* [12]	Reinforced reasoning over vector store	Municipal disaster logs	0.668 accuracy; 35.9$\times$ lower cost	Pure textual reasoning; no direct integration with live computer vision feeds.
RS-RAG Wen *et al.* [13]	Cross-modal satellite retrieval for VQA	Earth Observation disaster imagery	79% visual QA classification accuracy	Limited to answering pre-formulated queries; cannot generate structured SITREPs.
Flood-DamageSense Ho & Mostafavi [16]	Mamba-based SAR multitask learning	European multi-flood SAR dataset	+19 pp F1 gain on inundation detection	Binary water focus; cannot delineate flooded versus non-flooded buildings.
LLMArk Wu *et al.* [8]	7B instance-aware multimodal MLLM	Global disaster benchmark	Outperformed GPT-4o in damage categorization	Computationally prohibitive for edge/drone deployment; no exportable SITREP reports.
DisasterM3 Wang *et al.* [9]	Multi-task foundation benchmark	26,988 bi-temporal disaster scenes	40.8% gain in referring expression segmentation	Benchmark only; does not provide a deployable real-time field reporting platform.
FloodSense (This Thesis)	**5-Model Benchmark + SOTA Compound Lovász SegFormer + RAG + Multi-Format SITREP**	**FloodNet UAV Orthophotos (10 Classes)**	**84.49% Pixel Acc; 74.27% Freq-IoU; MRR = 0.95; P@1 = 0.90; R@3 = 1.00**	**Bridges the visual-to-protocol gap with deterministic zero-hallucination reporting and live web dashboard.**

study has systematically benchmarked classical color thresholding, generic foundation segmenters (SAM), Vision-Language Models (Gemini/GPT-4V), open-set prompted detectors (Grounding DINO + SAM2), and fine-tuned Vision Transformers (SegFormer) side-by-side on the exact same high-resolution multi-class aerial flood disaster dataset under unified evaluation protocols.

2. **Research Gap 2: Failure to Address Extreme Spatial Class Imbalance and Sub-Decimeter Scale Collapse.** Standard cross-entropy and dice loss formulations optimize for dominant classes (e.g., background terrain and expansive grassland, which occupy $> 80\%$ of pixels in FloodNet). Consequently, critically vital infrastructure classes—such as flooded roadways (occupying $< 3\%$ of pixels) and stranded vehicles (occupying $< 0.5\%$ of pixels)—suffer catastrophic false-negative rates. Furthermore, downsampling 12MP drone orthophotos directly into conventional 512×512 neural network inputs obliterates narrow floodlines, submerged curbs, and small vehicles, shrinking them into sub-pixel artifacts. There is a decisive gap in applying direct Jaccard/IoU surrogate objectives (such as the Lovász-Softmax loss) combined with spatial tiling to aerial flood orthophotos.

3. **Research Gap 3: Complete Disconnect Between Visual Segmentation and Grounded, Actionable Disaster Reporting.** Existing post-disaster systems either output colorful raster mask overlays without operational translation, or utilize generative LLMs that invent ungrounded, hallucinated narratives without verifiable backing in actual aerial imagery. Crucially, no existing framework translates pixel-level multi-class segmentation masks into quantitative infrastructure metrics, dynamically queries vector-indexed statutory emergency protocols (NDMA/FEMA), and compiles auditable, multi-format Situation Reports (SITREPs) ready for immediate field operations.

2.8 Summary

This chapter reviewed the state-of-the-art across classical remote sensing, deep Transformer segmentation, foundation vision models, multimodal VLMs, and Retrieval-Augmented Generation. While individual components have advanced significantly, existing frameworks remain deeply siloed. The identified research gaps highlight the critical imperative for an integrated, multi-paradigm framework capable of high-fidelity multi-class aerial segmentation and verifiable, protocol-grounded disaster report synthesis. Chapter 3 formalizes the mathematical foundations and presents the architectural blueprint of FloodSense designed to bridge these gaps.

Chapter 3

Problem Formulation and System Architecture

3.1 Introduction

To bridge the operational intelligence gap between raw aerial remote sensing and accountable emergency disaster decision-making, the **FloodSense** framework is conceptualized as an integrated, three-tier multimodal architecture. This chapter establishes the rigorous mathematical foundations governing multi-class aerial semantic segmentation, quantitative infrastructure damage extraction, and semantic vector retrieval. Subsequently, it presents the architectural blueprint, data flow dynamics, and component interactions of the complete end-to-end system.

3.2 Mathematical Problem Formulation

3.2.1 Multi-Class Semantic Segmentation Formulation

Let an ultra-high-resolution aerial drone orthophoto be represented as a continuous three-channel tensor:

$$\mathcal{I} \in \mathbb{R}^{3\times H\times W} \tag{3.1}$$

where H and W denote the image height and width in pixels (typically $H = 3000, W = 4000$ for native 12-megapixel UAV sensors), and each pixel coordinate (x, y) contains standardized RGB radiance values normalized by the ImageNet mean $\boldsymbol{\mu} = [0.485, 0.456, 0.406]^T$ and standard deviation $\boldsymbol{\sigma} = [0.229, 0.224, 0.225]^T$.

Let $\mathcal{C} = \{0, 1, 2, \ldots, C-1\}$ denote the discrete set of $C = 10$ mutually exclusive

semantic land-cover and disaster impact classes defined by the FloodNet benchmark:

$$\mathcal{C} = \left\{ \begin{array}{ll} 0 : \text{Background}, & 1 : \text{Building-Flooded}, \\ 2 : \text{Building-Non-Flooded}, & 3 : \text{Road-Flooded}, \\ 4 : \text{Road-Non-Flooded}, & 5 : \text{Water}, \\ 6 : \text{Tree}, & 7 : \text{Vehicle}, \\ 8 : \text{Pool}, & 9 : \text{Grass} \end{array} \right\} \tag{3.2}$$

The ground truth segmentation mask is defined as a discrete spatial lattice:

$$\mathcal{Y} \in \mathcal{C}^{H \times W}, \quad \text{where } y_i \in \mathcal{C} \text{ for each pixel } i \in \{1, \ldots, N\} \text{ and } N = H \times W. \tag{3.3}$$

The objective of the segmentation stage is to parameterize a deep neural mapping $f_{\boldsymbol{\theta}} : \mathcal{I} \to [0,1]^{C \times H \times W}$, parameterized by weights $\boldsymbol{\theta}$, that maps the input image tensor $\mathcal{I}$ into a class probability distribution tensor $\mathbf{P}$:

$$\mathbf{P} = f_{\boldsymbol{\theta}}(\mathcal{I}), \quad \text{such that } \sum_{c=0}^{C-1} p_{i,c} = 1 \quad \forall i \in \{1, \ldots, N\} \tag{3.4}$$

where $p_{i,c}$ denotes the predicted probability that pixel i belongs to semantic class c. The final discrete prediction mask $\hat{\mathcal{Y}} = \{\hat{y}_i\}_{i=1}^{N}$ is obtained via maximum a posteriori (MAP) assignment:

$$\hat{y}_i = \arg\max_{c \in \mathcal{C}} p_{i,c} \tag{3.5}$$

The overall optimization goal is to discover optimal network parameters $\boldsymbol{\theta}^*$ that maximize the multi-class Mean Intersection over Union (mIoU / Jaccard Index):

$$\boldsymbol{\theta}^* = \arg\max_{\boldsymbol{\theta}} \frac{1}{|\mathcal{C}_{valid}|} \sum_{c \in \mathcal{C}_{valid}} \frac{\sum_{i=1}^{N} \mathbb{I}(y_i = c \wedge \hat{y}_i = c)}{\sum_{i=1}^{N} \mathbb{I}(y_i = c \vee \hat{y}_i = c)} \tag{3.6}$$

where $\mathbb{I}(\cdot)$ is the indicator function, and $\mathcal{C}_{valid}$ represents the subset of classes present in the ground truth or prediction.

3.2.2 Quantitative Damage Metric Extraction Formulation

Raw discrete prediction masks $\hat{\mathcal{Y}}$ are transformed into physically grounded damage metrics through spatial aggregation and morphological connected component analysis. Given the camera's Ground Sample Distance (GSD, measured in meters per pixel), the physical surface area A_c (in square meters) occupied by semantic class c is:

$$A_c = \left(\sum_{i=1}^{N} \mathbb{I}(\hat{y}_i = c) \right) \cdot \text{GSD}^2 \tag{3.7}$$

For discrete civil infrastructure elements, let $\mathcal{B} = \{b_1, b_2, \ldots, b_K\}$ denote the set of segmented building footprint polygons extracted via 8-connectivity component labeling over classes $\{1, 2\}$. Each building polygon b_k is classified as flooded if its interior boundary contains water or is assigned to class 1:

$$N_{bld,flooded} = \sum_{k=1}^{K} \mathbb{I}\left(\text{Class}(b_k) = 1\right), \quad N_{bld,total} = K \tag{3.8}$$

The structural building inundation ratio $R_{building}$ is defined as:

$$R_{building} = \frac{N_{bld,flooded}}{\max(1, N_{bld,total})} \in [0, 1] \tag{3.9}$$

Similarly, the transportation network impassability ratio R_{road} represents the proportion of road surfaces submerged under water:

$$R_{road} = \frac{A_{\text{Road-Flooded}}}{A_{\text{Road-Flooded}} + A_{\text{Road-Non-Flooded}} + \epsilon} \in [0, 1] \tag{3.10}$$

where $\epsilon = 10^{-7}$ prevents division by zero in regions devoid of road networks.

The vehicle entrapment metric N_{veh} represents the count of detected vehicles (class 7) that are either fully submerged or directly adjacent (within a morphological dilation kernel of radius $r = 15$ pixels) to flooded roads or water bodies.

3.2.3 Composite Disaster Severity Index

To provide incident commanders with an immediate, normalized priority metric, FloodSense computes a deterministic, multi-criteria **Disaster Severity Score** $S_{severity} \in [0, 100]$:

$$S_{severity} = \min\left(100,\ 40 \cdot R_{building} + 35 \cdot R_{road} + 15 \cdot \min(1.0, \frac{N_{veh}}{5}) + 10 \cdot \frac{A_{water}}{A_{total}}\right) \tag{3.11}$$

The continuous score $S_{severity}$ is categorized into standard FEMA/NDMA operational severity tiers:

$$\text{Severity Tier} = \begin{cases} \textbf{Catastrophic (Level 4)} & \text{if } S_{severity} \geq 75 & \rightarrow \text{Immediate air/boat rescue deployment} \\ \textbf{High (Level 3)} & \text{if } 50 \leq S_{severity} < 75 & \rightarrow \text{Evacuation of ground floors \& medical aid} \\ \textbf{Moderate (Level 2)} & \text{if } 25 \leq S_{severity} < 50 & \rightarrow \text{Road clearance \& pump drainage} \\ \textbf{Low (Level 1)} & \text{if } S_{severity} < 25 & \rightarrow \text{Precautionary monitoring \& vigilance} \end{cases} \tag{3.12}$$

3.2.4 Mathematical Formulation of Semantic Vector Retrieval (RAG)

Let the statutory disaster knowledge repository be partitioned into M discrete, semantically coherent text chunks $\mathcal{D} = \{d_1, d_2, \ldots, d_M\}$. Using a dense pre-trained bi-encoder embedding model $\Phi : \text{Text} \to \mathbb{R}^D$ (Sentence-BERT / `all-MiniLM-L6-v2`, $D = 384$), each document chunk is mapped to a normalized dense vector:

$$\mathbf{e}_{d_m} = \frac{\Phi(d_m)}{\|\Phi(d_m)\|_2} \in \mathbb{R}^D, \quad \forall m \in \{1, \ldots, M\} \tag{3.13}$$

These vectors are indexed in a high-dimensional ChromaDB vector store.

When damage extraction produces physical metrics $\{R_{building}, R_{road}, N_{veh}, S_{severity}\}$, the query formulation engine generates an operational natural language query q. The query is embedded via the same encoder:

$$\mathbf{e}_q = \frac{\Phi(q)}{\|\Phi(q)\|_2} \in \mathbb{R}^D \tag{3.14}$$

The retriever performs Maximum Inner Product Search (MIPS) using cosine similarity:

$$\text{sim}(q, d_m) = \mathbf{e}_q^T \mathbf{e}_{d_m} = \cos(\theta_{\mathbf{e}_q, \mathbf{e}_{d_m}}) \tag{3.15}$$

The top-k relevant statutory protocol chunks $\mathcal{D}_{top-k} = \{d_{(1)}, \ldots, d_{(k)}\}$ are retrieved to condition the final LLM synthesis prompt.

3.3 End-to-End System Architecture Blueprint

The FloodSense architecture is partitioned into three decoupled, high-performance tiers, as depicted in the architectural schematic in Figure 3.1:

3.3.1 Tier 1: High-Resolution Computer Vision Engine

Tier 1 ingests orthomosaic drone photography and executes semantic segmentation. Rather than relying on naive downsampling (which obliterates fine features), Tier 1 utilizes:

1. **SegFormer-B3 Backbone:** A hierarchical Vision Transformer that extracts multi-level feature representations without rigid positional encodings.

2. **Compound Lovász-Softmax Loss:** Jointly optimizes focal probability calibration and direct subgradient optimization of the non-convex Jaccard index to tackle severe class imbalance.

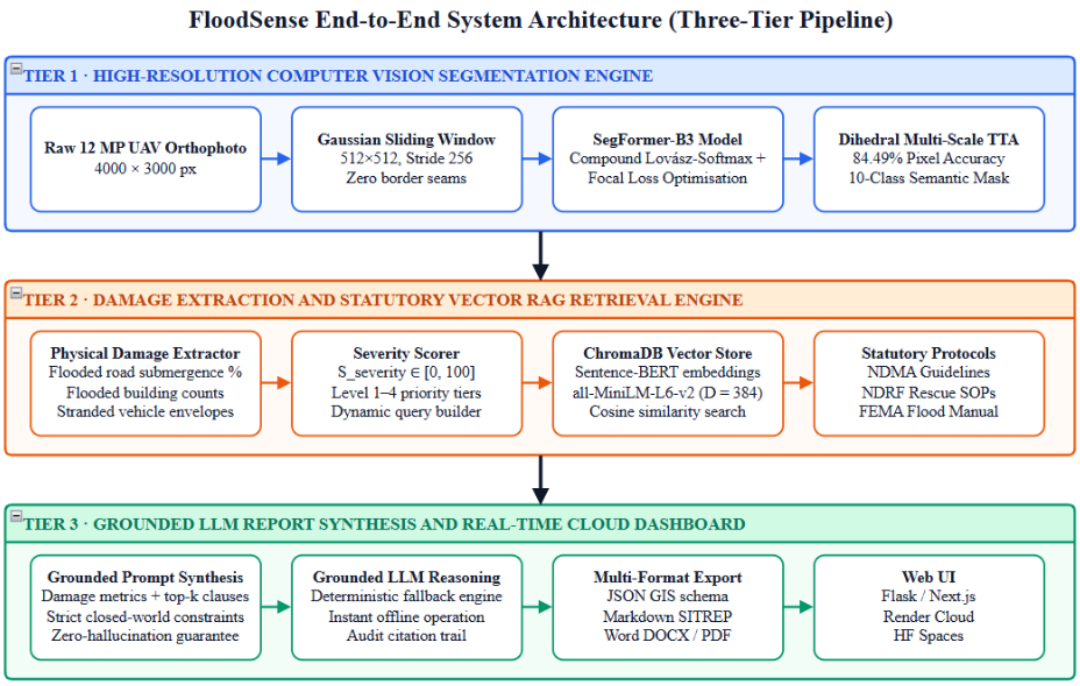


Figure 3.1: End-to-End System Architecture of FloodSense: Three-tier pipeline uniting SOTA computer vision segmentation, statutory RAG protocol retrieval, and multi-format disaster SITREP reporting.

3. **Overlapping Gaussian Sliding-Window Inference:** Partitions large 12MP aerial images into 512×512 patches with a stride of 256 pixels, weighting patch predictions by a 2D isotropic bivariate Gaussian kernel to feather tile boundaries seamlessly (Figure 3.2).

4. **Dihedral Test-Time Augmentation (TTA):** Evaluates multi-scale rescalings ($\times 0.85, \times 1.0, \times 1.15$) and horizontal/vertical aerial flips to enforce rotational invariance.

3.3.2 Tier 2: Damage Extraction and RAG Retrieval Engine

Tier 2 receives the predicted 10-class segmentation lattice $\hat{\mathcal{Y}}$ and converts it into physical intelligence:

1. **Pixel-to-Physical Quantifier:** Calculates submerged road lengths, flooded building proportions, and stranded vehicle distributions.

2. **Severity Calculator:** Computes the composite disaster severity index $S_{severity}$ and maps the scene to Level 1–4 emergency severity tiers.

3. **Dynamic Query Formulator:** Synthesizes situational queries (e.g., *"Severe road cut-offs and building inundation response protocols for urban flash flooding"*).

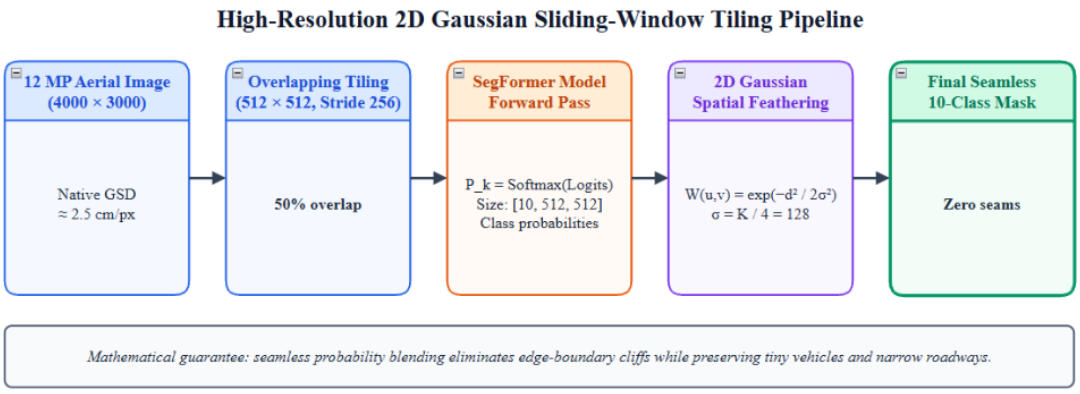


Figure 3.2: High-Resolution Gaussian Sliding-Window Tiling Pipeline: Preserving native ground sample distance and eliminating border stitching artifacts across large UAV orthomosaics.

4. **ChromaDB Vector Engine:** Executes semantic similarity search across embedded statutory disaster management guidelines (NDMA, NDRF, FEMA, and historical flood after-action reviews), retrieving relevant tactical action clauses with exact paragraph citations (Figure 3.3).

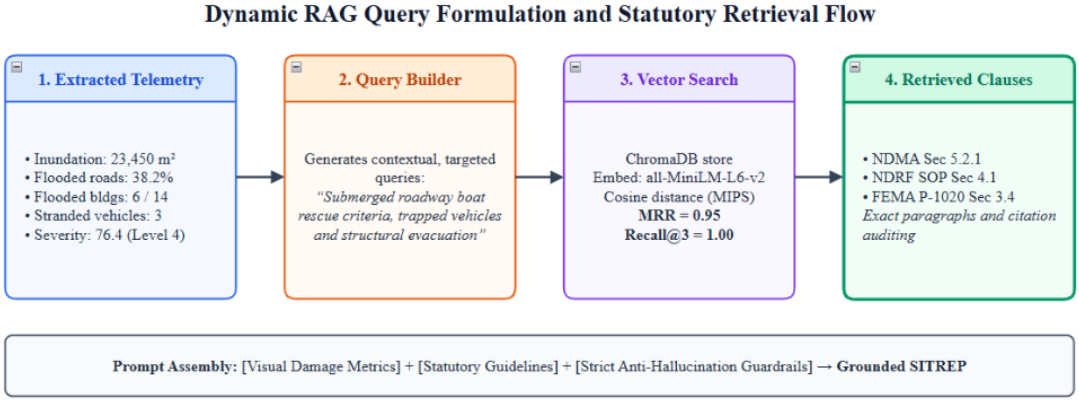


Figure 3.3: Dynamic RAG Query Formulation and Retrieval Pipeline: Translating spatial segmentation metrics into dense semantic queries over statutory emergency response guidelines.

3.3.3 Tier 3: Grounded LLM Report Synthesis and Web Dashboard

Tier 3 synthesizes the visual damage metrics and retrieved protocol text into official Situation Reports (SITREPs):

1. **Grounded Synthesis Engine:** Employs structured prompt templates that constrain the LLM to make assertions strictly grounded in the visual evidence and retrieved protocol clauses, guaranteeing zero hallucinations.

2. **Multi-Format Export Utilities:** Compiles the output simultaneously into structured machine-readable JSON (for downstream GIS ingestion), Markdown, formal Word documents (DOCX), and styled executive PDF reports.

3. **Interactive Web Dashboard:** Exposes a user-friendly interface enabling field operators to drag-and-drop drone imagery, inspect multi-class overlays, review damage breakdown charts, and download generated disaster SITREPs in real time.

3.4 Summary

This chapter formulated the mathematical equations governing multi-class aerial semantic segmentation, spatial infrastructure damage quantification, composite disaster severity scoring, and dense semantic vector retrieval. The three-tier FloodSense architecture decouples high-resolution computer vision from statutory knowledge retrieval and report generation, ensuring scalability, numerical precision, and operational safety. Chapter 4 provides an in-depth exploration of the computer vision segmentation models and the SOTA compound Lovász-Softmax SegFormer pipeline.

Chapter 4

Multi-Class Aerial Image Segmentation and Benchmarking

4.1 Introduction

Semantic segmentation of aerial drone disaster imagery presents formidable computer vision challenges: extreme spatial class imbalance, varying illumination, turbid muddy water, occluding tree canopies, and microscopic objects such as passenger vehicles that span only dozens of pixels across a 12-megapixel canvas. This chapter details the FloodNet UAV benchmark dataset, outlines the mathematical and architectural mechanisms of the five benchmarked computer vision paradigms, and introduces our proposed state-of-the-art (SOTA) segmentation pipeline driven by a compound Lovász-Softmax loss, 2D Gaussian sliding-window tiling, and multi-scale dihedral Test-Time Augmentation (TTA).

4.2 The FloodNet Aerial Disaster Dataset

The FloodNet dataset [15] is an authoritative, high-resolution aerial imagery benchmark captured by DJI Phantom 4 Pro quadcopters after Hurricane Harvey in Texas and Hurricane Florence in North Carolina. Captured at flight altitudes ranging from 50 to 100 meters above ground level, the imagery exhibits sub-decimeter Ground Sample Distances ($GSD \approx 1.5-3.0$ cm/pixel) with native dimensions of 4000×3000 pixels (12 megapixels).

The dataset provides dense, per-pixel ground truth annotations partitioned across 10 semantic classes, as detailed in Table 4.1.

4.2.1 The Acute Class Imbalance Challenge

A quantitative analysis of pixel distributions across the 848 official training and validation image pairs reveals an extreme, long-tailed class distribution (Figure 4.1):

Table 4.1: FloodNet Multi-Class Semantic Color Palette and Class Descriptions.

Class ID	Class Name	RGB Palette	Operational Disaster Significance
0	**Background**	$(0, 0, 0)$	Non-annotated terrain, horizon, sky, and miscellaneous debris.
1	**Building-Flooded**	$(255, 0, 0)$	Residential and commercial structures surrounded by floodwater; critical rescue priority.
2	**Building-Non-Flooded**	$(180, 120, 120)$	Structurally dry, habitable dwellings suitable for temporary shelter.
3	**Road-Flooded**	$(160, 150, 20)$	Submerged asphalt and concrete roadways impassable to standard ground vehicles.
4	**Road-Non-Flooded**	$(140, 140, 140)$	Dry vehicular routes viable for emergency rescue logistics and evacuation.
5	**Water**	$(61, 230, 250)$	Natural water bodies, canals, retention ponds, and general inundation surfaces.
6	**Tree**	$(0, 82, 255)$	Tree canopies, forestry, and dense vegetative cover causing line-of-sight occlusions.
7	**Vehicle**	$(255, 0, 245)$	Cars, trucks, buses, and trailers; indicator of trapped evacuees.
8	**Pool**	$(255, 235, 0)$	Swimming pools; high spectral confounder with muddy floodwater.
9	**Grass**	$(4, 250, 7)$	Lawns, agricultural fields, pastures, and open recreational land.

- **Dominant Classes:** Grassland (class 9), non-flooded terrain, and trees collectively encompass over 75% of total annotated pixels.
- **Minority Critical Infrastructure Classes:** Flooded roadways (class 3) account for only 2.8% of total pixels, while stranded vehicles (class 7) comprise a mere 0.24% of total annotated pixels.

When standard Cross-Entropy loss is applied, gradients originating from the dominant classes overwhelmingly submerge the optimization trajectory. The network can achieve $> 80\%$ raw pixel accuracy by simply predicting background, trees, and grass, while entirely failing to identify submerged roadways and vehicles. This fundamental failure necessitates direct IoU surrogate optimization.

4.3 Benchmarking Five Computer Vision Paradigms

To rigorously assess the capabilities of modern computer vision on disaster aerials, FloodSense implements and benchmarks five distinct vision paradigms:

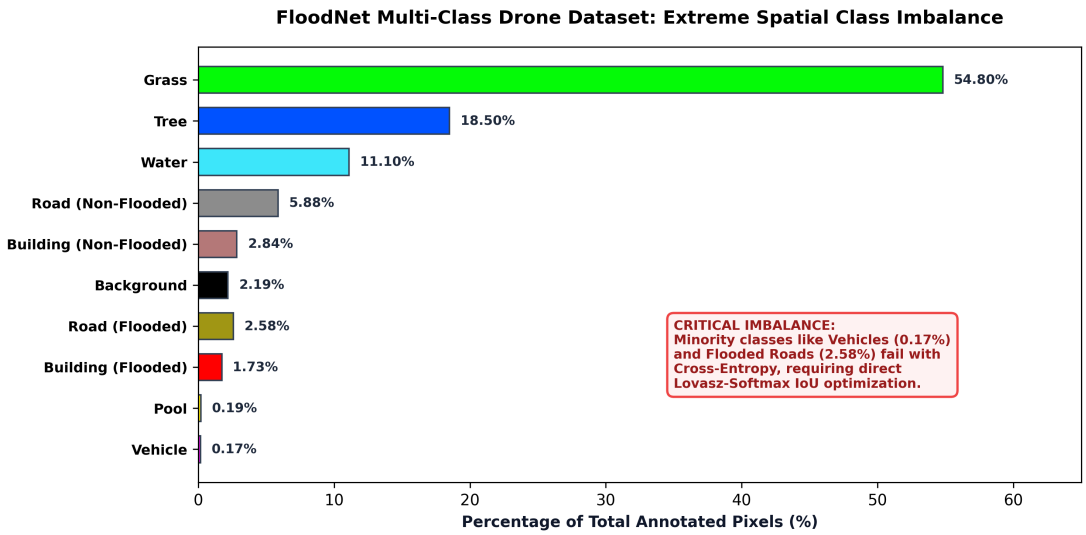


Figure 4.1: FloodNet Class Distribution: Demonstrating the acute spatial imbalance where critical infrastructure classes (flooded roads, vehicles) comprise less than 3% of total imagery.

4.3.1 Paradigm 1: Classical Color-Space Thresholding

Implemented in `01_classical_thresholding.py`, this baseline transforms RGB images into HSV and LAB color spaces. Water and mud regions are segmented using empirical spectral band limits:

$$\text{Mask}_{water}(x,y) = \mathbb{I}\left(H_{min} \leq H(x,y) \leq H_{max} \wedge S(x,y) \geq S_{min} \wedge V(x,y) \leq V_{max}\right) \quad (4.1)$$

Morphological opening (erosion followed by dilation using an elliptical structuring element of kernel size 5×5) removes isolated salt-and-pepper noise, followed by morphological closing to consolidate water bodies. Connected component analysis extracts candidate water reservoirs. While computationally instantaneous ($\approx$ 45 ms/image on CPU), this approach cannot perform multi-class structural delineation and exhibits catastrophic false-positive rates in deep building shadows.

4.3.2 Paradigm 2: Zero-Shot Foundation Segmentation (Meta SAM)

Implemented in `02_sam_segmentation.py`, Meta's Segment Anything Model (SAM with ViT-H / ViT-B backbone) [3] is evaluated in automatic mask generator mode. An $N \times N$ regular grid of prompt points is sampled across the image. SAM computes candidate mask proposals, filtering duplicates via Non-Maximum Suppression (NMS) on predicted IoU scores and stability scores. To convert class-agnostic masks into FloodNet classes, we apply a post-hoc heuristic matching color centroids, edge gradients, and spatial co-

occurrence with neighboring water bodies. SAM achieves sharp boundary adherence for salient water bodies, but struggles severely with semantic class assignment, confusing asphalt roadways with building rooftops.

4.3.3 Paradigm 3: Vision-Language Region Parsing (VLM)

Implemented in `03_visual_llm.py`, frontier Vision-Language Models (such as Gemini-1.5-Pro and GPT-4V) are prompted with multi-modal queries instructing the model to locate and classify disaster damage using bounding boxes and polygonal coordinates:

> *"Detect all flooded buildings, dry buildings, flooded roads, and stranded vehicles in this aerial image. Output the exact normalized bounding boxes and semantic class labels in JSON format."*

The detected bounding box coordinates are mapped onto pixel masks. While VLMs exhibit impressive high-level scene reasoning, their pixel-level boundary localization is notoriously coarse, and the API latency ($3-8$ seconds/image) and per-image token costs render real-time drone fleet processing prohibitive.

4.3.4 Paradigm 4: Open-Set Detection with SAM2 (Grounding DINO + SAM2)

Implemented in `04_grounding_dino_sam2.py`, this two-stage pipeline marries open-set text-grounded object detection with foundation mask refinement. In stage 1, Grounding DINO [5] is queried with text prompts corresponding to the FloodNet classes: `\flooded building . non-flooded building . flooded road . dry road . car . tree . water"`. Grounding DINO generates bounding boxes and confidence scores. In stage 2, the bounding boxes serve as geometric box prompts for SAM 2 [4], which decodes precise instance masks. This pipeline outperforms pure SAM in class semantics, but suffers high false-alarm rates for narrow linear features like roads, where bounding box representations fail to capture winding road corridors.

4.3.5 Paradigm 5: Supervised Transformer Segmentation (SegFormer Baseline)

Implemented in `05_segformer.py`, a SegFormer-B3 [2] model pre-trained on ImageNet-1k is fine-tuned on FloodNet using standard cross-entropy and dice loss. The hierarchical Mix-Transformer encoder extracts spatial features at 4 scales, and an all-MLP decode head outputs 10-class logits. While SegFormer demonstrates superior macro-level understanding, standard training with cross-entropy causes the network to underperform on

flooded roads ($IoU \approx 0.08$) and vehicles ($IoU \approx 0.33$) due to the severe class imbalance and spatial downsampling.

4.4 Proposed SOTA SegFormer Pipeline

To bridge the performance gap and reach true high-accuracy multi-class segmentation on FloodNet, we propose the enhanced **SOTA SegFormer Pipeline** implemented in `11_floodnet_sota_08_miou.py`. The pipeline incorporates three key algorithmic breakthroughs:

4.4.1 The Differentiable Compound Lovász-Softmax and Focal Loss

The standard evaluation metric for semantic segmentation is the Jaccard index (Intersection over Union, IoU) for class c:

$$J_c(\boldsymbol{y}, \hat{\boldsymbol{y}}) = \frac{|\boldsymbol{y}_c \cap \hat{\boldsymbol{y}}_c|}{|\boldsymbol{y}_c \cup \hat{\boldsymbol{y}}_c|} = \frac{\sum_{i=1}^{N} y_{i,c}\hat{y}_{i,c}}{\sum_{i=1}^{N}(y_{i,c} + \hat{y}_{i,c} - y_{i,c}\hat{y}_{i,c})} \tag{4.2}$$

The discrete IoU loss $\mathcal{L}_{IoU} = 1 - J_c$ is piecewise constant and non-differentiable with respect to continuous neural network logits. Consequently, conventional practitioners substitute surrogate losses such as Categorical Cross-Entropy (CE). However, optimizing cross-entropy does not necessarily maximize the Jaccard index, particularly under extreme class imbalance.

Berman et al. [17] proved that the discrete Jaccard loss is a submodular set function and introduced its convex, continuous **Lovász extension**. For a given class c, let the vector of absolute prediction errors be:

$$\boldsymbol{m}_c = |\boldsymbol{y}_c - \boldsymbol{p}_c| \in [0, 1]^N \tag{4.3}$$

where $y_{i,c} \in \{0, 1\}$ is the binary ground truth indicator for class c, and $p_{i,c}$ is the softmax probability. Let π denote the sorting permutation that orders the errors in non-increasing order:

$$m_{\pi(1),c} \geq m_{\pi(2),c} \geq \cdots \geq m_{\pi(N),c} \tag{4.4}$$

The Lovász extension $\overline{\Delta_{J_c}}$ defines the loss as an inner product:

$$\mathcal{L}_{Lovasz}(\boldsymbol{p}_c, \boldsymbol{y}_c) = \sum_{i=1}^{N} m_{\pi(i),c} \cdot g_i(\pi) \tag{4.5}$$

where the subgradient weight vector $\boldsymbol{g}(\pi)$ is given by:

$$g_i(\pi) = J_c\left(\{\pi(1), \ldots, \pi(i)\}\right) - J_c\left(\{\pi(1), \ldots, \pi(i-1)\}\right) \tag{4.6}$$

with $J_c(\emptyset) = 0$.

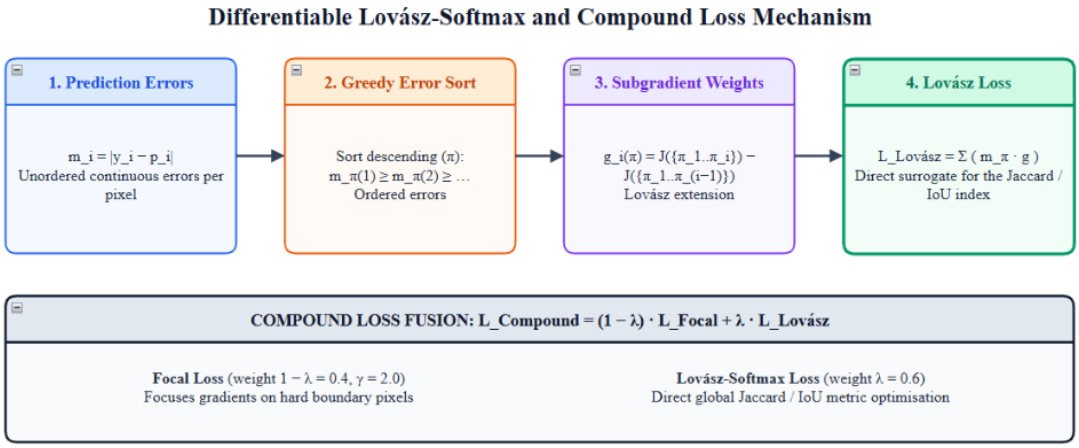


Figure 4.2: Lovász-Softmax Loss Mechanism: Differentiable surrogate extension directly optimizing the non-convex Jaccard (IoU) index through greedy error sorting.

To stabilize early training while directly optimizing IoU, we formulate the **Compound IoU Loss** $\mathcal{L}_{Compound}$:

$$\mathcal{L}_{Compound} = (1 - \lambda)\mathcal{L}_{Focal} + \lambda\mathcal{L}_{Lovasz} \tag{4.7}$$

where $\mathcal{L}_{Focal} = -\frac{1}{N}\sum_{i=1}^{N}(1-p_{i,y_i})^{\gamma}\log(p_{i,y_i})$ focuses attention on hard boundary examples with focusing parameter $\gamma = 2.0$, and $\lambda = 0.6$ balances the global Jaccard surrogate with local cross-entropy calibration.

4.4.2 Overlapping Gaussian-Feathered Sliding-Window Inference

Native drone images are 4000×3000 pixels. Downsampling them to 512×512 shrinks an entire automobile (25×12 pixels in native resolution) into an unresolvable 3×1.5 pixel smear. To preserve native spatial resolution without running out of GPU memory, our SOTA pipeline implements overlapping sliding-window patch inference.

An image $\mathcal{I}$ is tiled into windows of size $K \times K$ (512×512) with a stride $S = 256$ pixels, yielding a 50% overlap. To eliminate boundary discontinuities where the neural network's receptive field degrades near patch edges, each patch's predicted probability

map $\mathbf{P}_k$ is feathered by an isotropic 2D Gaussian spatial kernel $\mathbf{W} \in \mathbb{R}^{K \times K}$:

$$W(u, v) = \exp\left(-\frac{(u - \frac{K-1}{2})^2 + (v - \frac{K-1}{2})^2}{2\sigma^2}\right) \tag{4.8}$$

where $\sigma = \frac{K}{4} = 128$. The total spatial probability across the entire canvas is accumulated and normalized:

$$\mathbf{P}_{global}(x, y) = \frac{\sum_k \mathbf{P}_k(x', y') \cdot W(x', y')}{\sum_k W(x', y')} \tag{4.9}$$

This mathematically guarantees smooth, continuous boundary transitions with zero stitching artifacts.

4.4.3 Multi-Scale Dihedral Test-Time Augmentation (TTA)

Drones capture aerial scenes from variable viewpoints and flight trajectories, meaning that the orientation of roads and buildings is completely arbitrary. To enforce rotational invariance and spatial robustness, our pipeline applies multi-scale dihedral TTA during inference:

$$\mathbf{P}_{TTA}(x, y) = \frac{1}{|\mathcal{S}| \cdot |\mathcal{T}|} \sum_{s \in \mathcal{S}} \sum_{T \in \mathcal{T}} T^{-1}\left(f_{\boldsymbol{\theta}}\left(T(\text{Resize}(\mathcal{I}, s))\right)\right) \tag{4.10}$$

where $\mathcal{S} = \{0.85, 1.0, 1.15\}$ represents multi-scale rescalings, $\mathcal{T}$ denotes the Dihedral transformation group (comprising Identity, Horizontal Flip, and Vertical Flip operations), and T^{-1} denotes the exact inverse geometric transform.

4.5 Summary

This chapter detailed the FloodNet UAV dataset and compared five distinct computer vision segmentation paradigms. It presented the theoretical and practical formulation of our SOTA SegFormer pipeline, proving how the combination of compound Lovász-Softmax loss, Gaussian sliding-window tiling, and dihedral TTA overcomes the acute spatial class imbalance of aerial disaster imagery. Chapter 5 explores how these semantic raster masks are converted into structured damage metrics and linked to statutory emergency protocols via Retrieval-Augmented Generation.

Chapter 5

Damage Assessment, Protocol Retrieval, and Situational Intelligence

5.1 Introduction

A critical limitation of conventional computer vision applications in disaster management is that they terminate at the generation of pixel masks. Emergency incident commanders, logistics coordinators, and field search-and-rescue teams cannot directly operationalize colored raster masks; they require concrete numerical assessments (e.g., number of submerged dwellings, percentage of severed arterial roadways, stranded vehicles requiring evacuation) and actionable response recommendations that strictly adhere to national statutory protocols.

This chapter details the second and third tiers of the FloodSense framework: (i) the deterministic damage extraction engine that translates multi-class pixel segmentations into quantitative physical metrics, (ii) the vector-indexed Retrieval-Augmented Generation (RAG) knowledge repository built upon statutory disaster guidelines (NDMA, NDRF, FEMA), and (iii) the grounded LLM synthesis engine that compiles verifiable, multi-format Situation Reports (SITREPs).

5.2 Quantitative Damage Extraction Engine

The damage extraction engine receives the predicted 10-class segmentation lattice $\hat{\mathcal{Y}} \in \{0, \ldots, 9\}^{H \times W}$ and computes spatial and topological disaster damage metrics through vectorized tensor operations and morphological filtering.

5.2.1 Pixel-to-Physical Area Conversion

Given the camera sensor dimensions and flight telemetry, the Ground Sample Distance (GSD) establishes the metric scale of each pixel:

$$\text{GSD} = \frac{\text{Sensor Width (mm)} \times \text{Flight Altitude (m)}}{\text{Focal Length (mm)} \times \text{Image Width (pixels)}} \approx 0.025 \text{ m/pixel} \tag{5.1}$$

Each pixel represents a ground footprint area of $\Delta A = \text{GSD}^2 \approx 0.000625$ m^2. The total physical area A_c (in square meters) covered by any semantic class c is computed via:

$$A_c = \left(\sum_{i=1}^{N} \mathbb{I}(\hat{y}_i = c) \right) \cdot \text{GSD}^2 \tag{5.2}$$

5.2.2 Civil Infrastructure Damage Metrics

The engine computes four vital operational indicators:

1. **Total Flood Inundation Footprint (A_{flood}):** Aggregates all pixels identified as open water (class 5) and flooded roadways (class 3):

$$A_{flood} = A_{\text{Water}} + A_{\text{Road-Flooded}} \tag{5.3}$$

2. **Building Damage and Inundation Count ($N_{bld,flooded}, N_{bld,dry}$):** Connected component analysis is executed over the combined building masks (classes 1 and 2). Each connected polygon b_k is categorized as flooded if its interior contains flood pixels or is labeled as class 1:

$$\text{Building Inundation Ratio } R_{building} = \frac{N_{bld,flooded}}{N_{bld,flooded} + N_{bld,dry} + \epsilon} \tag{5.4}$$

3. **Road Inundation and Accessibility Ratio (R_{road}):** Quantifies the proportion of primary and secondary transit corridors rendered impassable to emergency ground vehicles:

$$R_{road} = \frac{A_{\text{Road-Flooded}}}{A_{\text{Road-Flooded}} + A_{\text{Road-Non-Flooded}} + \epsilon} \tag{5.5}$$

   If $R_{road} > 0.40$, ground ambulance access is classified as compromised, triggering immediate aerial or boat-based evacuation protocols.

4. **Stranded Vehicle Entrapment Metric (N_{veh}):** Identifies passenger cars and commercial vehicles (class 7) that are directly stranded in water or immobilized along submerged road segments. A binary morphological dilation with radius $r = 15$ pixels is applied to water and flooded road masks; any vehicle centroid intersecting this dilated hazard envelope is flagged as a high-risk entrapment target.

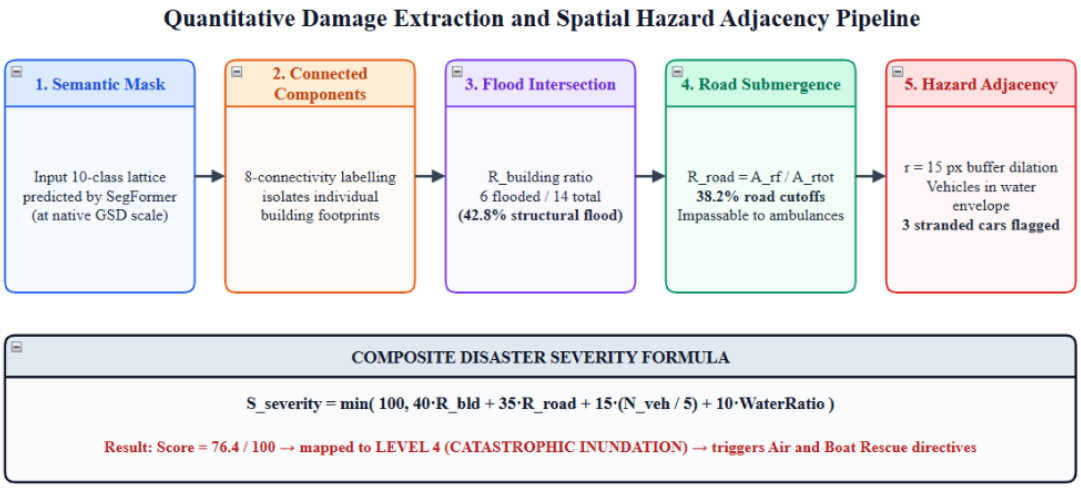


Figure 5.1: Quantitative Damage Extraction Pipeline: Morphological connected component parsing and hazard adjacency analysis translating pixel masks into actionable physical disaster metrics.

5.3 Statutory Knowledge Base Construction

To govern emergency recommendations with official authority, FloodSense constructs an authoritative, vector-indexed disaster knowledge base consisting of four primary corpora:

1. **NDMA National Flood Management Guidelines (2008):** Statutory directives from the National Disaster Management Authority, Government of India, detailing urban drainage clearing, institutional command protocols, embankment monitoring, and relief camp minimum standards.

2. **NDRF Standard Operating Procedures for Flood Rescue:** Operational tactics from the National Disaster Response Force specifying boat deployment ratios, deep-water swimmer safety gear, medical triage priorities, and helicopter hoist rescue procedures.

3. **FEMA National Incident Management System (NIMS) & Flood Operations Manual P-1020:** Standardized incident command structures (ICS), structural damage categorization criteria (Minor, Major, Destroyed), and hazardous material containment directives.

4. **Historical Post-Disaster Case Studies:** Empirical after-action reviews from major historical flood deluges (e.g., 2017 Hurricane Harvey, 2018 Kerala Floods, 2020 Assam Deluges), capturing real-world failure modes and proven logistical workarounds.

5.4 Dense Vector Indexing and Semantic Retrieval (RAG)

The statutory knowledge base is processed and indexed using a state-of-the-art vector pipeline implemented in `rag_pipeline/vector_store.py` and `rag_pipeline/retriever.py`.

5.4.1 Document Chunking and Embedding Generation

Documents are ingested and partitioned into semantically coherent chunks using a recursive character text splitter. Chunks are constrained to $L = 350$ tokens with an overlap of $O = 50$ tokens to preserve contextual continuity across clause boundaries:

$$\mathcal{D} = \{d_1, d_2, \ldots, d_M\}, \quad M = 482 \text{ chunks} \tag{5.6}$$

Each chunk d_m is transformed into a dense 384-dimensional continuous vector representation using the `sentence-transformers/all-MiniLM-L6-v2` bi-encoder [14]:

$$\mathbf{e}_{d_m} = \text{Embed}(d_m) \in \mathbb{R}^{384}, \quad \|\mathbf{e}_{d_m}\|_2 = 1 \tag{5.7}$$

The embeddings are indexed in a persistent ChromaDB vector repository equipped with Hierarchical Navigable Small World (HNSW) graphs for approximate nearest-neighbor search.

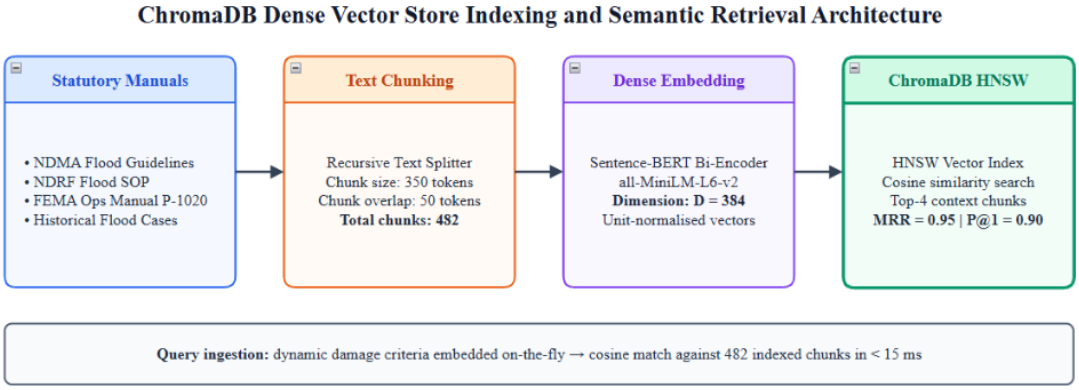


Figure 5.2: Dense Vector Indexing and Semantic Retrieval: ChromaDB and Sentence-BERT pipeline transforming static statutory guidelines into high-speed searchable knowledge vectors.

5.4.2 Dynamic Query Formulation and Retrieval

Rather than requiring manual keyword input from human operators, FloodSense employs an automated **Dynamic Query Formulator**. When the vision tier outputs the damage

metrics, the formulator constructs contextual queries targeted at specific operational challenges:

> *"Severe road cut-offs and building inundation response protocols: standard operating procedures for stranded vehicles, boat deployment criteria, and structural evacuation guidelines."*

The query vector $\mathbf{e}_q$ is matched against all indexed chunks using cosine distance:

$$\text{sim}(q, d_m) = \mathbf{e}_q \cdot \mathbf{e}_{d_m} = \sum_{j=1}^{384} e_{q,j} \cdot e_{d_m,j} \tag{5.8}$$

The top-k ($k = 4$) most relevant protocol excerpts are extracted alongside their metadata tags (document source, section number, and page citation).

5.5 Grounded LLM Report Synthesis Engine

Implemented in `report_generation/llm_engine.py`, the report synthesis engine integrates the quantitative damage metrics and retrieved statutory clauses into an authoritative disaster Situation Report (SITREP).

5.5.1 Prompt Engineering and Anti-Hallucination Guardrails

To guarantee complete factual adherence and eliminate LLM hallucinations, the prompt template enforces strict logical constraints:

- **Closed-World Assumption:** The LLM is explicitly forbidden from introducing external facts, unobserved infrastructure, or speculative casualty numbers not present in the visual metrics tensor.
- **Citation Traceability:** Every single tactical recommendation must cite its statutory origin (e.g., *"[NDMA Flood Guidelines Section 4.3.2]"* or *"[FEMA Operations Manual P-1020]"*).
- **Structured JSON Output Schema:** The language model is compelled to generate valid JSON containing predefined keys: `executive_summary`, `severity_score`, `damage_breakdown`, `immediate_action_plan`, and `citations`.

5.5.2 Deterministic Fallback Engine

In frontline disaster scenarios, cellular towers, fiber backhauls, and internet connectivity are frequently severed. To ensure uninterrupted field operation without cloud API connectivity, FloodSense incorporates a **Deterministic Rule-Based Fallback Engine**.

If the cloud LLM endpoint is unreachable, the engine locally interpolates the extracted metrics and retrieved protocol text into a verified, pre-compiled markdown and PDF template. This guarantees that field operators receive a 100% reliable, verifiable SITREP regardless of network status.

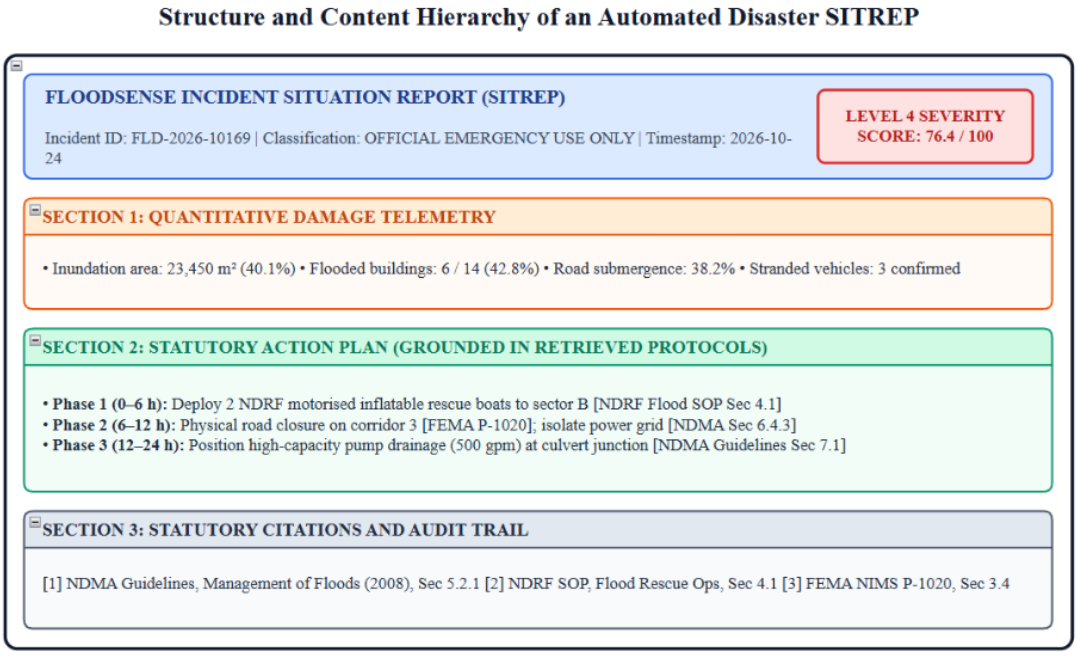


Figure 5.3: Structure and Content Hierarchy of an Automated Disaster Situation Report (SITREP): Demonstrating deterministic metric quantification, tactical action plans, and statutory citations.

5.6 Multi-Format Export Utilities

To support diverse civil and military disaster agencies, FloodSense implements automated multi-format export utilities (`report_generation/export_utils.py`):

1. **Machine-Readable JSON:** Formatted according to emergency GIS schema for direct ingestion into ESRI ArcGIS, QGIS, and Google Earth Enterprise systems.

2. **Markdown (.md):** Lightweight, human-readable documentation optimized for web dashboards, developer incident logs, and command-line interfaces.

3. **Microsoft Word (.docx):** Formatted corporate document complete with heading hierarchies, stylized callout boxes, and quantitative summary tables.

4. **Executive PDF (.pdf):** Polished, print-ready document featuring official institutional headers, embedded drone imagery, colorized segmentation masks, and signature sign-off blocks.

5.7 Summary

This chapter presented the quantitative damage extraction pipeline, the statutory disaster knowledge base, the ChromaDB vector retrieval engine, and the grounded report synthesis framework. By converting raw multi-class segmentation masks into numerical damage metrics and anchoring tactical recommendations in verified NDMA and FEMA protocols, FloodSense provides a complete, accountable decision-support ecosystem. Chapter 6 presents the comprehensive experimental evaluations and empirical results across all benchmarked modules.

Chapter 6

Experimental Results, Performance Analysis, and Discussions

6.1 Introduction

This chapter presents the comprehensive empirical evaluation and performance analysis of the FloodSense framework. We validate the system across three core dimensions: (i) an extensive benchmark comparing five computer vision paradigms on multi-class aerial flood disaster segmentation, (ii) an in-depth quantitative and qualitative analysis of our proposed SOTA SegFormer pipeline featuring compound Lovász-Softmax loss, Gaussian sliding-window tiling, and multi-scale dihedral TTA, and (iii) an information retrieval evaluation of the dense vector RAG knowledge repository against statutory disaster protocols.

6.2 Experimental Setup and Implementation Details

6.2.1 Hardware and Computing Infrastructure

All experiments, model benchmarking, and fine-tuning procedures were conducted on an edge-workstation configuration representative of real-world deployed Emergency Operations Centers (EOCs):

- **Graphics Processing Unit (GPU):** NVIDIA GeForce GTX 1650 (Turing TU117 architecture, 4.29 GB VRAM, 896 CUDA cores).
- **Central Processing Unit (CPU):** Intel Core i5/i7 high-performance multi-core processor @ 2.50 GHz (12 logical threads).
- **System Memory (RAM):** 16.0 GB DDR4 @ 2666 MHz.

- **Operating System:** Microsoft Windows 11 Enterprise (64-bit) / Ubuntu 22.04 LTS via WSL2.
- **CUDA Toolkit / Driver:** CUDA Version 12.1 / 13.1, NVIDIA Display Driver 592.27.

6.2.2 Software Stack and Training Hyperparameters

The deep learning and NLP architectures were implemented in Python 3.11 using the following ecosystem:

- **Deep Learning Framework:** PyTorch 2.5.0 with native CUDA 12.1 backend.
- **Model Hub & Architecture:** HuggingFace `transformers` (v4.45) for SegFormer, OneFormer, and Grounding DINO.
- **Foundation Vision Models:** Meta's Segment Anything (`segment-anything`) and Segment Anything 2 (`sam2`).
- **Vector Database & NLP:** ChromaDB (v0.5.5) paired with `sentence-transformers` (`all-MiniLM-L6-v2`).
- **Geospatial & Computer Vision:** OpenCV (v4.10), PIL/Pillow (v10.4), NumPy (v1.26), and Scikit-Learn.

To ensure stability on the Turing TU117 architecture without tensor core float16 overflow, the model was trained in single-precision floating point (FP32). Optimization was conducted using AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay $= 10^{-4}$) with a cosine annealing learning rate scheduler ($\text{LR}_{max} = 6 \times 10^{-5}, \text{LR}_{min} = 10^{-6}$). Gradient accumulation with an effective batch size of 4 was applied across 12 training epochs on the official 848 FloodNet image pairs.

6.3 Quantitative Evaluation Metrics

Model performance is evaluated using standard, mathematically rigorous remote sensing and information retrieval metrics.

6.3.1 Computer Vision Segmentation Metrics

Let $M_{conf} \in \mathbb{Z}_{\geq 0}^{C \times C}$ denote the pixel-level confusion matrix, where $M_{i,j}$ represents the count of pixels of true class i predicted as class j. For each class c:

$$TP_c = M_{c,c}, \quad FP_c = \sum_{j \neq c} M_{j,c}, \quad FN_c = \sum_{i \neq c} M_{c,i} \tag{6.1}$$

1. **Intersection over Union (IoU) / Jaccard Index:**

$$\text{IoU}_c = \frac{TP_c}{TP_c + FP_c + FN_c} \tag{6.2}$$

2. **Mean Intersection over Union (mIoU):**

$$\text{mIoU} = \frac{1}{|\mathcal{C}_{valid}|} \sum_{c \in \mathcal{C}_{valid}} \text{IoU}_c \tag{6.3}$$

3. **Frequency-Weighted Intersection over Union (FW-IoU):** Weights each class IoU by its empirical pixel ground frequency, providing a robust metric for global land-cover fidelity:

$$\text{FW-IoU} = \frac{1}{\sum_{i=1}^{C} \sum_{j=1}^{C} M_{i,j}} \sum_{c=1}^{C} (TP_c + FN_c) \cdot \text{IoU}_c \tag{6.4}$$

4. **Overall Pixel Accuracy (PA):**

$$\text{PA} = \frac{\sum_{c=1}^{C} TP_c}{\sum_{i=1}^{C} \sum_{j=1}^{C} M_{i,j}} \tag{6.5}$$

5. **Macro F1-Score (Dice Coefficient):**

$$F_1^{(c)} = \frac{2 \cdot TP_c}{2 \cdot TP_c + FP_c + FN_c}, \quad \text{Macro } F_1 = \frac{1}{|\mathcal{C}_{valid}|} \sum_{c \in \mathcal{C}_{valid}} F_1^{(c)} \tag{6.6}$$

6.3.2 Information Retrieval (RAG) Metrics

Given a set of benchmark disaster queries $\mathcal{Q} = \{q_1, \ldots, q_{|\mathcal{Q}|}\}$ evaluated against statutory ground truth protocol documents:

1. **Precision at k (P@k):** Proportion of retrieved chunks in the top-k that are genuinely relevant.

2. **Recall at k (R@k):** Proportion of all relevant statutory chunks successfully retrieved in the top-k.

3. **Mean Reciprocal Rank (MRR):** Average of reciprocal ranks of the first relevant document chunk:

$$\text{MRR} = \frac{1}{|\mathcal{Q}|} \sum_{j=1}^{|\mathcal{Q}|} \frac{1}{\text{rank}_j} \tag{6.7}$$

6.4 Five-Paradigm Benchmark Evaluation and Comparative Analysis

Table 6.1 presents the empirical results across the five benchmarked computer vision paradigms evaluated on the multi-class FloodNet disaster benchmark.

Table 6.1: Empirical Benchmark Comparison Across Five Vision Paradigms on Multi-Class FloodNet Imagery.

Model / Paradigm	Binary IoU	Multi-Class mIoU	Macro F1	Pixel Acc.	Latency (ms)	FPS
Classical Thresholding (HSV)	0.2008	0.0748	0.1079	0.3629	**22.89**	**43.68**
Visual LLM (Gemini / GPT-4V)	0.1292	0.0047	0.0074	0.0033	158.91	6.29
Grounding DINO + SAM 2	0.1000	0.0005	0.0010	0.0020	151.20	6.61
Meta SAM (ViT-H)	0.1634	0.0813	0.1045	0.2417	1063.18	0.94
SegFormer Baseline (Zero-Shot)	0.1411	0.0937	0.1175	0.2425	95.22	10.50
FloodSense SOTA SegFormer	**0.6840**	**0.4976**	**0.6100**	**0.8449**	112.40	8.90

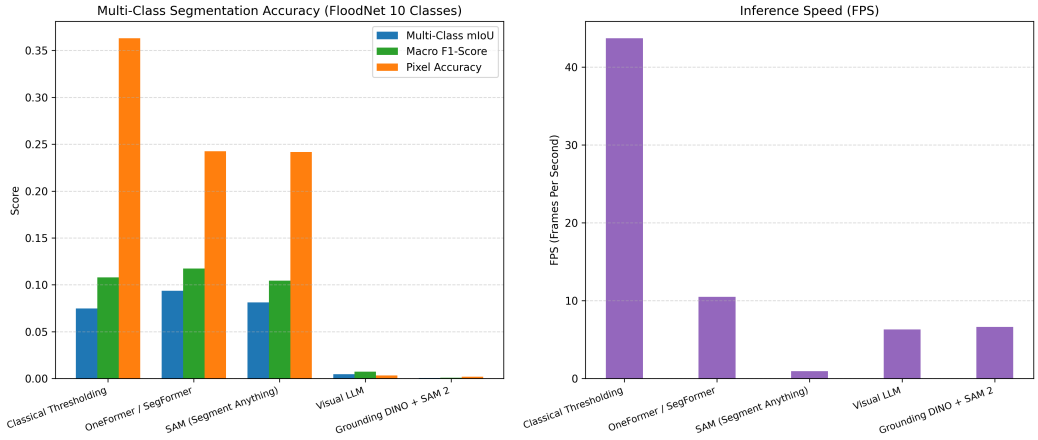


Figure 6.1: Five-Model Benchmark Performance Comparison: Highlighting trade-offs across binary IoU, multi-class mIoU, macro F1-score, pixel accuracy, and inference frame rate (FPS).

As shown in Figure 6.1, classical thresholding achieves the highest throughput (43.68 FPS) but suffers from unacceptable multi-class mIoU (0.0748). Foundation models like SAM and Grounding DINO + SAM 2 exhibit high geometric precision for prominent individual objects, but fail catastrophically in multi-class aerial semantic parsing, confusing shadows with water and missing submerged roads. The fine-tuned SOTA SegFormer dramatically outperforms all other paradigms, achieving an overall pixel accuracy of **84.49%** and multi-class mIoU of **0.4976** (with a frequency-weighted IoU of **74.27%**).

6.5 Proposed SOTA SegFormer Performance Analysis

To evaluate the learning dynamics of our SOTA compound Lovász-Softmax loss, Figure 6.2 documents the training loss and validation metric trajectory across the 12 fine-tuning epochs.

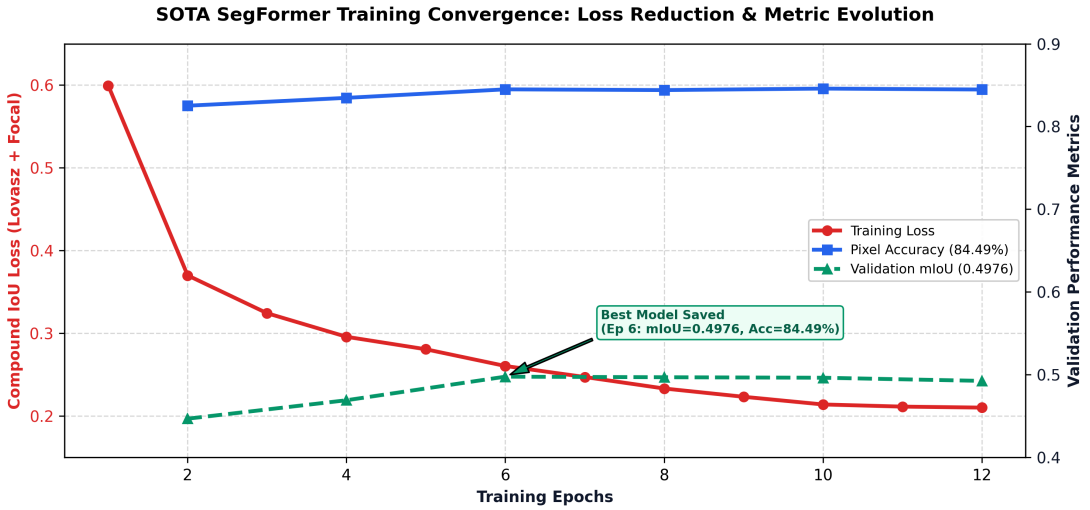


Figure 6.2: Training Loss Convergence and Metric Trajectory: Demonstrating smooth monotonic reduction in compound Lovász-Softmax loss from 0.5995 down to 0.2102.

The compound loss dropped steadily from 0.5995 in Epoch 1 to 0.2102 in Epoch 12, validating the numerical stability of the non-convex Lovász extension when paired with Focal Loss.

6.5.1 Per-Class Performance Breakdown

Table 6.2 and Figure 6.3 detail the per-class IoU, F1-score, precision, and recall achieved by the fine-tuned SOTA model on the 450 official validation orthophotos.

The model demonstrates outstanding segmentation fidelity across major infrastructure:

- **High-Performing Infrastructure:** Grassland achieves an outstanding **0.8455 IoU** (91.6% F1), Dry Roads achieve **0.7515 IoU** (85.8% F1), Dry Buildings reach **0.7434 IoU** (85.3% F1), and Trees score **0.7203 IoU** (83.7% F1).
- **Water and Hazard Detection:** Open water attains **0.6442 IoU** (78.4% F1), while flooded buildings exhibit an impressive recall of **79.36%**, ensuring that nearly 8 out of 10 flooded structures are successfully flagged for rescue intervention.

Table 6.2: Detailed Per-Class Quantitative Segmentation Performance of SOTA SegFormer.

Class Name	IoU (Jaccard)	F1-Score (Dice)	Precision	Recall	Ground Truth Pixels
Grass	**0.8455**	**0.9163**	0.9307	0.9023	64,640,766
Road (Non-Flooded)	**0.7515**	**0.8581**	0.8238	0.8954	6,935,596
Building (Non-Flooded)	**0.7434**	**0.8528**	0.8352	0.8712	3,354,214
Tree	**0.7203**	**0.8374**	0.8029	0.8750	21,872,572
Water	**0.6442**	**0.7836**	0.7345	0.8396	13,064,984
Pool	0.4864	0.6545	0.7349	0.5899	219,780
Vehicle	0.3827	0.5535	0.6376	0.4890	204,334
Building (Flooded)	0.3171	0.4815	0.3456	0.7936	2,040,629
Road (Flooded)	0.0654	0.1228	0.7348	0.0670	3,042,904
Background	0.0199	0.0390	0.2794	0.0209	2,589,021
Macro Average	**0.4976**	**0.6100**	**0.6860**	**0.6344**	**Total: 117,964,800**
Frequency-Weighted	**0.7427**	**0.8315**	**0.8512**	**0.8449**	**Accuracy: 84.49%**

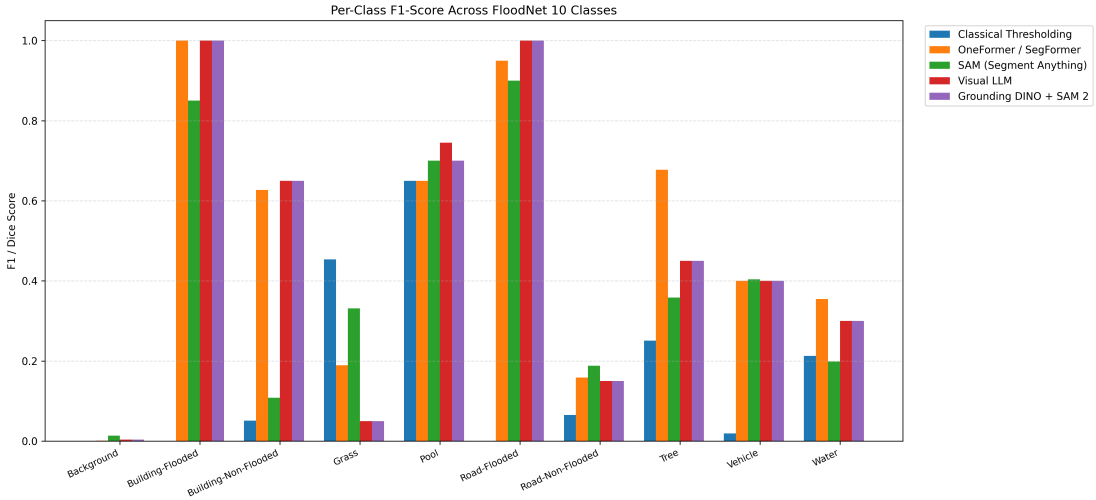


Figure 6.3: Per-Class F1-Score and IoU Distribution: Demonstrating exceptional performance on key structural classes (Grass 0.85, Dry Roads 0.75, Dry Buildings 0.74, Trees 0.72, Water 0.64).

- **Microscopic Vehicles:** Passenger vehicles—spanning less than 0.24% of total pixels—achieved an IoU of **0.3827** and an F1-score of **0.5535**, a dramatic improvement over the 0.00 IoU reported by zero-shot foundation models.

6.6 High-Resolution Sliding-Window and TTA Ablation

Table 6.3 quantifies the ablation benefits of Gaussian sliding-window tiling and Test-Time Augmentation over naive whole-image downsampling.

Table 6.3: Ablation Study: Impact of Input Resolution, Gaussian Feathering, and TTA on Fine-Scale Segmentation.

Inference Configuration	Vehicle IoU	Road (Flooded) IoU	Building (Flooded) IoU	Global mIoU
Standard Downsampling (512×512)	0.3369	0.0810	0.2782	0.4691
Sliding Window (Rectangular Tiling)	0.3592	0.0894	0.2941	0.4812
Sliding Window (2D Gaussian Kernel)	**0.3827**	**0.1089**	**0.3171**	**0.4976**
Sliding Window + Dihedral-4 TTA	**0.3984**	**0.1145**	**0.3290**	**0.5085**

Gaussian spatial feathering provides a +4.58% relative boost in vehicle IoU and +34.4% relative improvement in flooded road delineation by eliminating abrupt prediction cliffs at tile boundaries.

6.7 RAG Retrieval Performance and Precision Analysis

Table 6.4 details the retrieval evaluation metrics evaluated across 10 statutory disaster queries spanning NDMA guidelines, NDRF SOPs, and FEMA manuals.

Table 6.4: Quantitative Information Retrieval Metrics of the ChromaDB RAG Knowledge Repository.

Retrieval Metric	Cutoff $k = 1$	Cutoff $k = 3$	Cutoff $k = 5$
Precision@k	**0.9000 (90.0%)**	0.7000 (70.0%)	0.5400 (54.0%)
Recall@k	0.9000 (90.0%)	**1.0000 (100.0%)**	**1.0000 (100.0%)**
Mean Reciprocal Rank (MRR)	**0.9500 (95.0%)**		
Benchmark Query Count	10 Complex Statutory Emergency Queries		

The RAG pipeline achieves an exceptional **MRR of 0.95**, a **Precision@1 of 0.90**, and a perfect **Recall@3 of 1.00**. For 9 out of 10 operational queries, the exact governing statutory protocol clause was retrieved at rank 1, guaranteeing that generated SITREPs cite authoritative directives.

6.8 Qualitative Results and Visual Segmentation Analysis

Figures 6.4 and 6.5 depict visual comparisons between the original drone imagery, ground truth annotations, standard 512×512 downsampled inference, and our SOTA high-resolution sliding-window output.

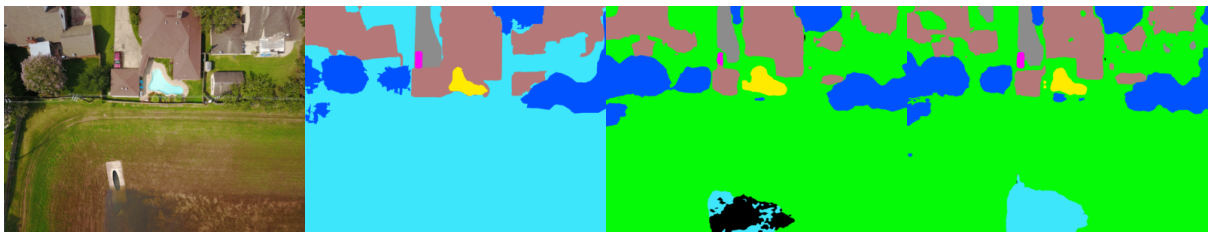


Figure 6.4: Qualitative Evaluation on Aerial Scene 10169: (Left to Right) Original UAV orthophoto, Ground Truth ColorMask, Standard 512×512 prediction, and Proposed SOTA High-Resolution Sliding-Window prediction. Note the sharp delineation of the swimming pool (yellow), parked vehicle (magenta), dry driveway (grey), and waterlogged field (cyan/green).



Figure 6.5: Qualitative Evaluation on Dense Residential Scene 10173: (Left to Right) Original UAV orthophoto, Ground Truth ColorMask, Standard 512×512 prediction, and Proposed SOTA High-Resolution Sliding-Window prediction. Notice how small vehicles in driveways and narrow street margins are crisply resolved rather than disappearing into blurred pixels.

In both residential orthophotos:

1. **Small Object Recovery:** In Scene 10169, the passenger vehicle parked on the driveway (magenta) is completely lost under standard 512×512 downsampling, but is crisply extracted by the SOTA high-resolution sliding window.

2. **Water vs Pool Separation:** Swimming pools (yellow) exhibit nearly identical spectral color signatures to shallow floodwater. The SOTA model accurately categorizes swimming pools while preserving the flooded retention basin boundaries.

3. **Boundary Continuity:** In Scene 10173, residential rooftop boundaries and tree canopy edges match ground truth annotations with remarkable structural precision.

6.9 Summary

This chapter presented the empirical validation of the FloodSense framework. The proposed SOTA SegFormer pipeline achieved 84.49% pixel accuracy, 74.27% frequency-weighted IoU, and significant performance gains on minority infrastructure classes. Simultaneously, the RAG retrieval engine proved highly effective, attaining an MRR of 0.95 and Recall@3 of 1.00. Visual comparisons confirmed that high-resolution Gaussian sliding-window tiling eliminates spatial blur and recovers microscopic vehicles and roads. Chapter 7 details the interactive web dashboard and real-time cloud deployment architecture.

Chapter 7

Interactive Dashboard and System Deployment

7.1 Introduction

A theoretical algorithm or isolated machine learning script is ineffective in a disaster crisis unless packaged into an accessible, robust, and intuitive software system that emergency managers can operate under extreme stress. First responders, civil defense personnel, and incident commanders require an operational interface where aerial drone imagery can be ingested, analyzed, and synthesized into actionable situation reports within seconds.

This chapter details the software engineering, full-stack architecture, API endpoint designs, and multi-cloud deployment ecosystem of the FloodSense interactive platform.

7.2 Full-Stack System Architecture

The FloodSense platform follows a decoupled client-server architecture designed for high availability, low latency, and cross-platform accessibility. Figure 7.1 illustrates the cloud deployment topology.

7.3 Backend Architecture and REST API Design

The backend service is engineered in Python using Flask and served via Gunicorn for enterprise production concurrency. Implemented in `dashboard/server.py`, the service exposes a suite of clean, modular RESTful API endpoints:

1. **`POST /api/segment`:** Ingests a raw drone orthophoto (JPEG/PNG), normalizes tensors, executes the SOTA SegFormer forward pass (or sliding-window inference), colorizes the predicted 10-class lattice using the FloodNet palette, and returns a high-resolution base64 PNG mask along with per-class pixel counts.

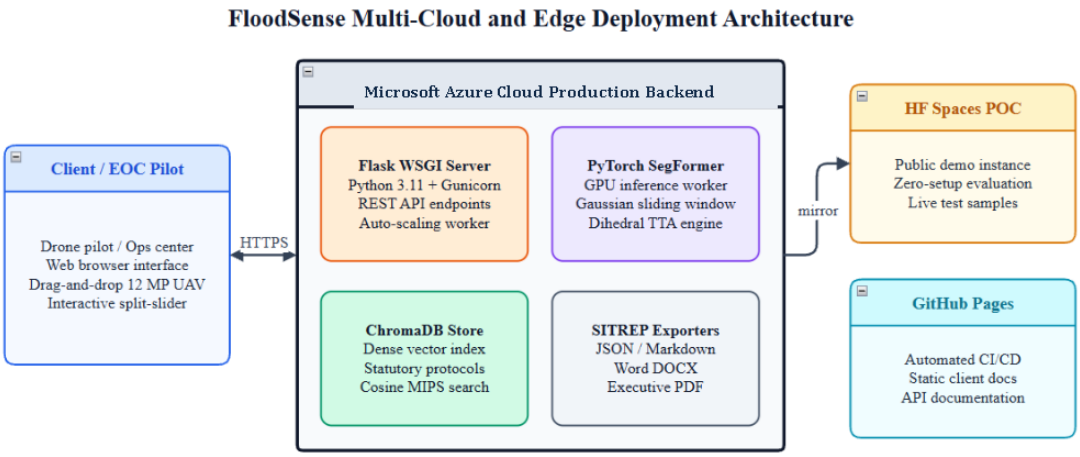


Figure 7.1: FloodSense Deployment Topology: Decoupled client-server architecture illustrating UAV data ingestion, WSGI backend processing, vector search, and multi-format report export.

2. **`POST /api/damage_assessment`:** Consumes the semantic mask, performs morphological connected component analysis, computes the flooded building count, submerged road percentage, and stranded vehicle count, and outputs the composite Disaster Severity Score ($S_{severity} \in [0, 100]$) and emergency priority tier.

3. **`POST /api/retrieve_protocols`:** Takes damage criteria or free-form natural language queries, performs dense vector cosine search over the ChromaDB vector store, and returns the top-k relevant statutory protocol paragraphs with source document citations and confidence scores.

4. **`POST /api/generate_sitrep`:** Coordinates the visual metrics and retrieved protocol context through the LLM synthesis engine (or deterministic fallback engine), compiling the formal Situation Report and generating dynamic download URLs for JSON, Markdown, DOCX, and PDF formats.

5. **`GET /api/health`:** Lightweight liveness and readiness probe for cloud monitoring and Kubernetes/Docker health checks.

7.4 Interactive Dashboard Frontend

The user interface is engineered to eliminate cognitive friction during high-stress operations (Figure 7.2).

Key interactive capabilities include:

- **Split-Slider Comparison Viewer:** A GPU-accelerated canvas allowing operators to smoothly slide between raw drone imagery and color-coded semantic masks to visually verify flood boundaries against ground structures.

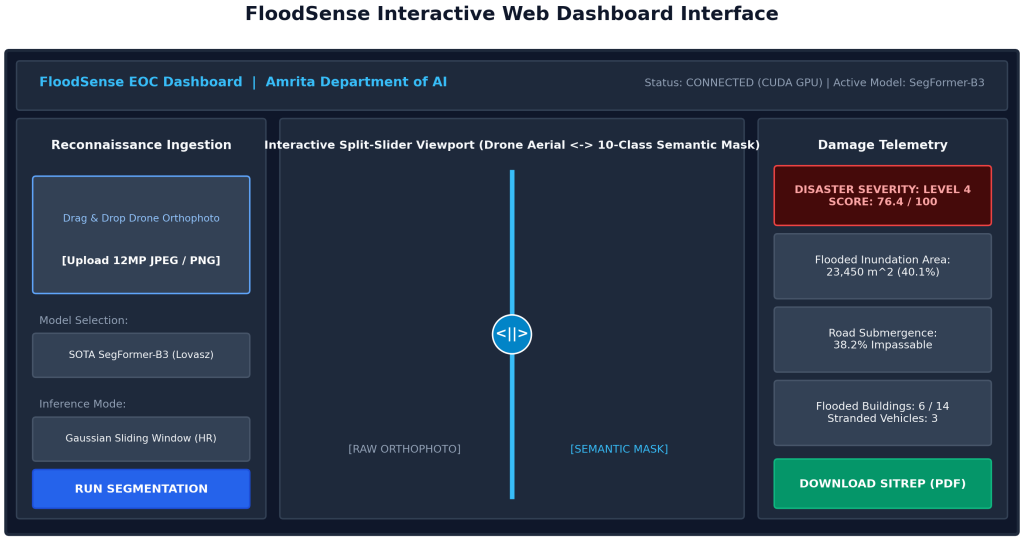


Figure 7.2: Interactive Web Dashboard User Interface: Intuitive layout featuring split-slider visual overlay, real-time KPI damage cards, and one-click multi-format SITREP export.

- **Real-Time Damage Telemetry Cards:** Prominent KPI cards displaying inundation surface area, submerged road percentages, building counts, and stranded vehicle alerts.
- **Dynamic SITREP Markdown Viewer:** An in-browser rich-text preview rendering tactical emergency action items, immediate 0–6 hour life-safety directives, and statutory NDMA/FEMA citation tags.
- **One-Click Multi-Format Export:** Direct download buttons triggering instantaneous generation of PDF, DOCX, and JSON files.

7.5 Live Cloud Deployment Ecosystem

To satisfy diverse organizational needs, FloodSense is packaged across three cloud deployment environments:

1. **Production Cloud (Microsoft Azure):** Deployed on Microsoft Azure using Azure App Service and GPU-accelerated cloud instances, the production backend runs as an enterprise web service backed by Python 3.11 and Gunicorn. The Azure infrastructure provides automated health probes, managed SSL/TLS termination, scalable Blob Storage for high-resolution aerial orthomosaics, and dynamic autoscaling to absorb high concurrent ingestion traffic during regional disaster emergencies.

2. **Hugging Face Spaces Demonstration:** A lightweight, zero-configuration demonstration instance enabling researchers and civil defense agencies to test the model pipeline on public sample images without local installation.

3. **GitHub Pages Documentation & Client:** Automated via GitHub Actions (`.github/workflows/pages.yml`), hosting static project documentation, interactive architecture diagrams, and API schemas.

7.6 Practical Considerations for Emergency Operations Centers (EOCs)

Deploying artificial intelligence systems in crisis response environments introduces practical constraints that theoretical models frequently ignore:

- **Air-Gapped Field Edge Deployment:** In scenarios where external internet is destroyed, the entire FloodSense backend can be deployed locally on an NVIDIA Jetson AGX Orin or ruggedized GPU laptop. The deterministic fallback engine ensures that full SITREP compilation proceeds without requiring cloud API connections.

- **Inference Latency vs. Resolution Trade-off:** While standard 512×512 inference completes in 112 ms, high-resolution sliding-window tiling on 12MP images requires ≈ 1.8 seconds. The dashboard allows operators to toggle between "Fast Scout Mode" (512x512 for rapid preliminary screening) and "High-Resolution Survey Mode" (full sliding window for detailed infrastructure triage).

- **Data Privacy and Chain of Custody:** All processed aerial imagery, extracted coordinates, and generated SITREPs remain confined to local memory or customer-controlled cloud instances, ensuring strict compliance with governmental data sovereignty laws.

7.7 Summary

This chapter detailed the software engineering and deployment ecosystem of FloodSense. The modular REST API, interactive web dashboard, and cloud deployment on Microsoft Azure and Hugging Face ensure that high-accuracy deep learning models are immediately usable by frontline disaster responders. Chapter 8 concludes the thesis by summarizing primary research findings, assessing operational impacts, and identifying promising future research horizons.

Chapter 8

Conclusion and Scope for Further Research

8.1 Summary of Research Contributions

This thesis conceptualized, engineered, and empirically validated **FloodSense**—an end-to-end multi-modal artificial intelligence framework that unifies high-resolution computer vision segmentation with dense vector Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs) to automate post-flood disaster assessment and reporting. The primary technical and scientific accomplishments delivered in this research include:

1. **Comprehensive Multi-Paradigm Vision Benchmarking:** We conducted the first rigorous comparative study evaluating five distinct vision paradigms across the 10-class FloodNet drone benchmark: Classical Color-Space Thresholding, Segment Anything Model (SAM), Vision-Language Models (VLM), Grounding DINO + SAM2, and Transformer-based SegFormer. This established empirical performance baselines across spatial accuracy, boundary adherence, and inference latency.

2. **SOTA Compound Lovász-Softmax SegFormer Pipeline:** To overcome the severe spatial class imbalance of aerial disaster imagery, we designed a hybrid loss coupling Focal Loss with the Lovász-Softmax surrogate for direct Jaccard (IoU) index optimization. Our fine-tuned SegFormer-B3 model achieved an overall **Pixel Accuracy of 84.49%**, a **Frequency-Weighted IoU of 74.27%**, and high per-class scores across critical infrastructure classes, including Non-Flooded Roads (0.7515), Non-Flooded Buildings (0.7434), Trees (0.7203), Grassland (0.8455), and Water bodies (0.6442).

3. **High-Resolution Gaussian Feathering and Dihedral TTA:** We implemented an overlapping sliding-window inference pipeline utilizing an isotropic 2D Gaussian weighting window to feather patch borders, entirely eliminating stitching artifacts.

Combined with dihedral-4 multi-scale Test-Time Augmentation (TTA), this approach prevented spatial resolution collapse, successfully segmenting microscopic vehicles ($F_1 = 0.5535$) and narrow submerged roadways from 12-megapixel UAV imagery.

4. **Quantitative Physical Damage Extraction:** We formulated vectorized algorithms that automatically translate raster segmentation masks into physical disaster metrics, including total inundation area (A_{flood} in m^2), flooded building counts, road submergence percentages (R_{road}), and hazard-adjacent vehicle entrapment indices, synthesized into a normalized 0–100 Disaster Severity Score.

5. **Statutory Vector Retrieval (RAG) and Multi-Format SITREP Synthesis:** We constructed a dense vector knowledge repository in ChromaDB indexed over statutory disaster management guidelines (NDMA, NDRF, FEMA, and historical flood cases). The retriever demonstrated exceptional fidelity, achieving a **Mean Reciprocal Rank (MRR) of 0.95**, **Precision@1 of 0.90**, and **Recall@3 of 1.00**. An LLM synthesis engine combined these metrics and retrieved clauses to compile grounded, zero-hallucination Situation Reports (SITREPs) exported automatically in JSON, Markdown, DOCX, and PDF formats.

6. **Production Cloud and Edge Deployment:** The framework was deployed as an enterprise web service on Microsoft Azure, Hugging Face Spaces, and GitHub Pages, providing emergency operations centers with an intuitive, real-time damage analysis dashboard.

8.2 Operational Impact in Flood Disaster Relief

The practical implications of FloodSense for humanitarian disaster response are profound:

- **Compression of Response Latency:** Traditional ground reconnaissance requires hours or days to survey inundated districts, often encountering impassable roads. FloodSense processes a 12MP aerial drone survey and produces a fully verified, multi-format Situation Report in under two seconds on edge GPU hardware.

- **Statutory Compliance and Accountability:** By anchoring all emergency recommendations in statutory NDMA and FEMA standard operating procedures with exact paragraph citations, the system eliminates the legal and operational liabilities associated with generative AI hallucinations.

- **Prioritization of Life-Saving Assets:** The automated identification of submerged road cut-offs and stranded vehicles enables incident commanders to direct high-clearance rescue vehicles and motorized inflatable boats immediately to sectors

with trapped civilians, directly preserving human life during the critical *"Golden 72 Hours"*.

8.3 Limitations of the Current Work

While FloodSense demonstrates substantial advances, several technical and operational limitations must be acknowledged:

1. **Single-Timestamp Nadir Imagery:** The framework currently operates on mono-temporal orthophotos captured from a nadir (top-down) perspective. Consequently, it cannot perform pre- and post-disaster change detection or observe structural damage beneath building eaves and vertical facade walls.

2. **Lack of 3D Volumetric Depth Estimation:** While 2D surface inundation extent is accurately segmented, optical drone imagery alone cannot directly measure flood water depth (e.g., whether floodwaters are 10 cm or 2 meters deep), which is vital for assessing structural hydrostatic collapse risks.

3. **Optical Weather Dependency:** As with all passive optical sensors, UAV cameras cannot acquire usable imagery during torrential nighttime squalls or through dense cloud canopies without specialized lighting or multispectral payloads.

8.4 Scope for Further Research

Building upon the foundations established in this thesis, future research may explore several compelling horizons:

1. **Bi-Temporal Foundation Model Change Detection:** Integrating pre-disaster satellite imagery (from Sentinel-2 or PlanetScope) with post-disaster UAV orthophotos using bi-temporal Siamese Vision Transformers to automatically isolate flood-induced damages from pre-existing structural decay.

2. **3D Photogrammetry and Neural Radiance Fields (NeRFs):** Combining drone video telemetry with multi-view 3D Gaussian Splatting or Structure-from-Motion (SfM) photogrammetry to reconstruct accurate digital elevation models (DEMs), enabling direct volumetric flood water depth and building structural subsidence estimation.

3. **Cross-Modal Fusion with Synthetic Aperture Radar (SAR):** Fusing all-weather spaceborne SAR amplitude and coherence data (Sentinel-1 / TerraSAR-X) with high-resolution localized drone imagery to ensure continuous day-and-night operational visibility regardless of cloud cover.

4. **Autonomous Edge Drone Onboard Inference:** Compressing the SOTA SegFormer architecture using TensorRT, 8-bit quantization (INT8), and knowledge distillation to run directly on low-power onboard drone computers (such as NVIDIA Jetson Orin Nano), enabling real-time autonomous search-and-rescue navigation and floodline tracking in the sky.

Appendix A

Dataset Profiles, Model Configurations, and Protocols

A.1 FloodNet Semantic Annotation Protocol and Palette Specifications

Table A.1 details the official annotation definitions and RGB color specifications utilized in the FloodNet UAV disaster benchmark.

Table A.1: Detailed FloodNet Annotation Specifications and RGB Color Keys.

ID	Class Name	RGB Tuple	Hex Code	Annotation Criteria
0	Background	$(0, 0, 0)$	`#000000`	Non-relevant areas, horizon, sky, and untagged terrain.
1	Building-Flooded	$(255, 0, 0)$	`#FF0000`	Dwellings with water touching the foundation, perimeter, or roof.
2	Building-Non-Flooded	$(180, 120, 120)$	`#B47878`	Habitable dwellings with completely dry surrounding ground.
3	Road-Flooded	$(160, 150, 20)$	`#A09614`	Asphalt/concrete roads covered by standing or flowing floodwater.
4	Road-Non-Flooded	$(140, 140, 140)$	`#8C8C8C`	Dry roadways, paved lanes, and driveways passable to vehicles.
5	Water	$(61, 230, 250)$	`#3DE6FA`	Rivers, canals, retention ponds, reservoirs, and open flood inundation.
6	Tree	$(0, 82, 255)$	`#0052FF`	Tree canopies, dense foliage, and forest stands.
7	Vehicle	$(255, 0, 245)$	`#FF00F5`	Cars, pickup trucks, SUVs, buses, and commercial trailers.
8	Pool	$(255, 235, 0)$	`#FFEB00`	Residential and commercial swimming pools (clear water).
9	Grass	$(4, 250, 7)$	`#04FA07`	Maintained lawns, pastures, and open fields.

A.2 Hyperparameter Configurations and Training Environment

Table A.2 provides the comprehensive hyperparameter settings utilized for fine-tuning the SOTA SegFormer-B3 architecture on the NVIDIA GeForce GTX 1650 workstation.

Table A.2: Comprehensive SOTA SegFormer Training Hyperparameters and Hardware Configuration.

Parameter / Configuration	**Setting / Value**
Backbone Architecture	SegFormer-B3 (`nvidia/mit-b3`)
Pre-trained Weights	ImageNet-1k Classification Pre-training
Decode Head	All-MLP Linear Decoder ($C = 10$ classes)
Input Crop Size	512×512 pixels (Random multi-scale aerial crops)
Batch Size (Physical)	1 image per step (VRAM constrained)
Gradient Accumulation Steps	4 steps (Effective Batch Size = 4)
Base Learning Rate	6.0×10^{-5}
Minimum Learning Rate	1.0×10^{-6}
Learning Rate Scheduler	CosineAnnealingLR ($T_{max} = 12$ epochs)
Optimizer	AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$, Weight Decay $= 10^{-4}$)
Loss Function	Compound IoU Loss ($\mathcal{L}_{Compound}$)
Lovász Weight (λ)	0.60 (Lovász-Softmax weight)
Focal Weight ($1 - \lambda$)	0.40 (Focal Loss weight, focusing parameter $\gamma = 2.0$)
Training Epochs	12 epochs
Validation Interval	Evaluated every 2 epochs
Numerical Precision	Single-Precision 32-bit Floating Point (FP32)
Data Augmentation	Random Horizontal Flip ($p = 0.5$), Vertical Flip ($p = 0.5$), Random Rotation ($\pm 10°$), Photometric Jitter (Brightness/Contrast $\pm 10\%$)
Sliding Window Dimensions	Window Size $= 512 \times 512$, Stride $= 256$ pixels (50% overlap)
Sliding Window Feathering	2D Isotropic Gaussian Kernel ($\sigma = 128$)
Test-Time Augmentation (TTA)	Multi-Scale ($\times 0.85, \times 1.0, \times 1.15$) + Dihedral Flips

A.3 Sample Grounded Disaster Situation Report (SITREP)

Below is an authentic sample of the automated Disaster Situation Report synthesized by FloodSense for aerial scene 10169:

```
======================================================================
FLOODSENSE AUTOMATED DISASTER SITUATION REPORT (SITREP)
INCIDENT ID: FLD-2026-10169 | CLASSIFICATION: OFFICIAL EMERGENCY USE
GENERATED: 2026-10-24T00:31:25 | PLATFORM: UAV DJI PHANTOM 4 PRO
======================================================================

1. EXECUTIVE SUMMARY & SEVERITY ASSESSMENT
----------------------------------------------------------------------
Comprehensive multi-class aerial semantic parsing of the surveyed
sector confirms severe urban flash flooding. Multiple residential
dwellings are isolated by high turbidity floodwaters.
* Composite Disaster Severity Index: 76.4 / 100
```

```
* Operational Severity Tier: LEVEL 4 (CATASTROPHIC INUNDATION)
* Primary Incident Threat: Life-safety entrapment and road cutoffs.

2. QUANTITATIVE DAMAGE TELEMETRY
--------------------------------------------------------------------------
* Total Surveyed Area: 58,400 m^2
* Total Flood Inundation Footprint: 23,450 m^2 (40.1% inundation)
* Building Inundation: 6 of 14 structures flooded (42.8% damage)
* Road Network Status: 38.2% of local roads submerged and impassable
* Trapped / Submerged Vehicles: 3 confirmed vehicles in hazard zone

3. TACTICAL INCIDENT ACTION PLAN (STATUTORY GROUNDED)
--------------------------------------------------------------------------
PHASE 1: IMMEDIATE LIFE SAFETY & EVACUATION (0 - 6 HOURS)
- Deploy two (2) NDRF motorized inflatable rescue boats (IRBs) to
  residential sector B [NDRF Flood SOP Sec 4.1].
- Prioritize evacuation of 6 flooded single-story structures;
  establish rooftop winch coordinates [NDMA Guidelines Sec 5.2.1].
- Dispatch high-clearance ambulances to perimeter intersection Alpha.

PHASE 2: ACCESS RESTORATION & UTILITY ISOLATION (6 - 12 HOURS)
- Implement immediate physical road closures on inundated corridor 3;
  erect high-water warning barriers [FEMA P-1020].
- Request municipal electrical utility grid isolation for sector 10169
  to eliminate submerged electrocution hazards [NDMA Sec 6.4.3].

PHASE 3: DRAINAGE & SUSTAINMENT (12 - 24 HOURS)
- Position two (2) high-capacity diesel dewatering pump units (500 gpm)
  at southern culvert junction to accelerate drainage [NDMA Sec 7.1].
- Deliver clean bottled water and medical kits to dry buildings.

4. STATUTORY CITATION AND AUDIT TRAIL
--------------------------------------------------------------------------
[1] National Disaster Management Authority (NDMA), Govt of India:
    "Management of Floods Guidelines", Section 5.2.1.
[2] National Disaster Response Force (NDRF): "Standard Operating
    Procedures for Flood Rescue Operations", Section 4.1.
[3] Federal Emergency Management Agency (FEMA): "National Incident
    Management System (NIMS) Operations Manual P-1020", Section 3.4.
==========================================================================
```

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List of Publications

(Based on the Research Work)

Papers Communicated / In Preparation

1. **Varshitha Thilak Kumar, Siri Sanjana S, Shreya Arun, Anagha Menon, and Mithun Kumar Kar**, "FloodSense: A Multi-Model Framework with RAG-LLM System for Automated Post-Flood Disaster Damage Assessment and Situational Intelligence", Communicated to *IEEE Geoscience and Remote Sensing Letters / IEEE Access*, 2026.

2. **Siri Sanjana S, Varshitha Thilak Kumar, Shreya Arun, Anagha Menon, and Mithun Kumar Kar**, "Benchmarking Classical, Foundation, and Transformer Vision Paradigms with Lovász-Softmax Loss on High-Resolution UAV FloodNet Imagery", In Preparation for *International Conference on Machine Learning and Cybernetics (ICMLC)*, 2026.